

BFS Traversal

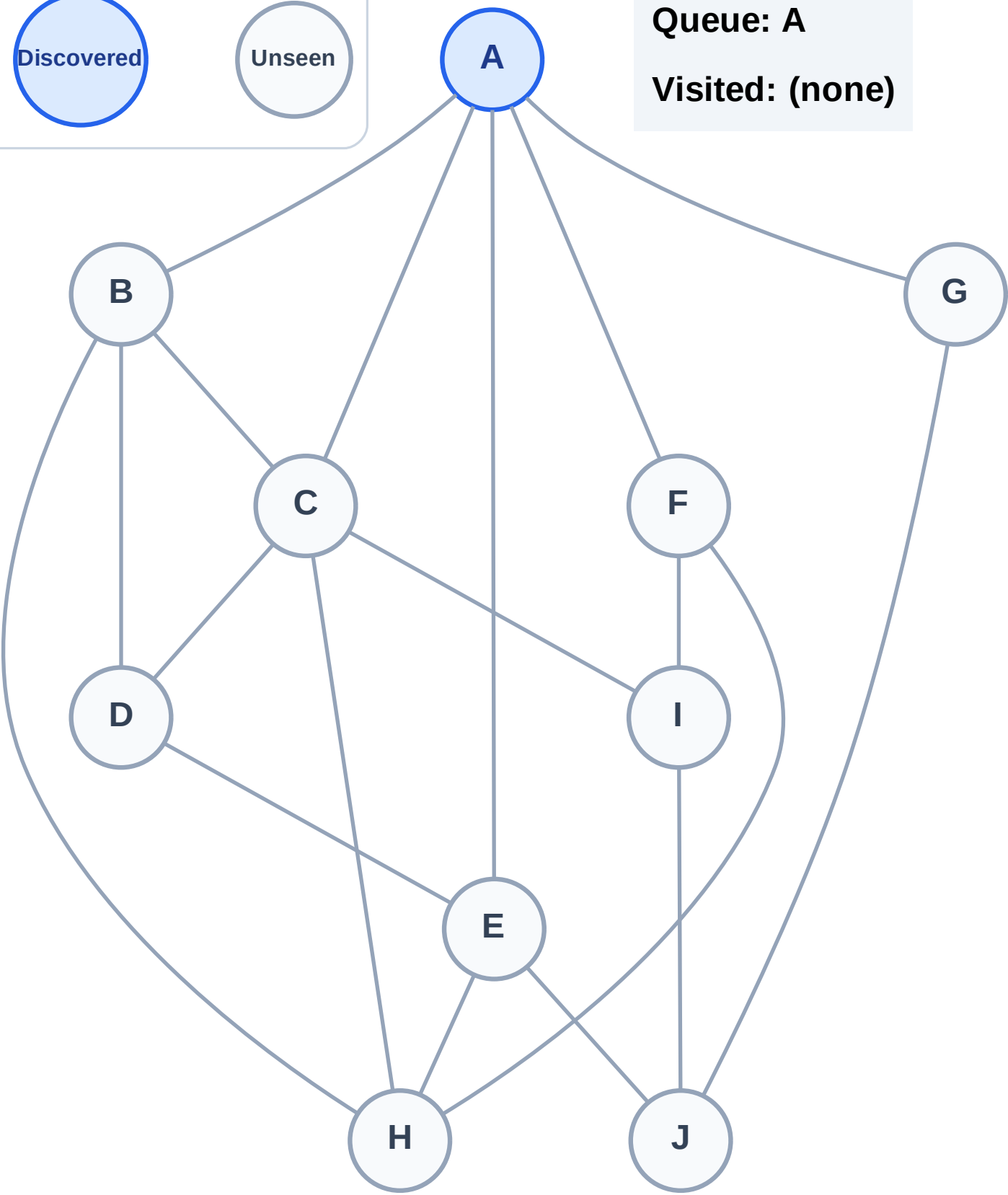
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

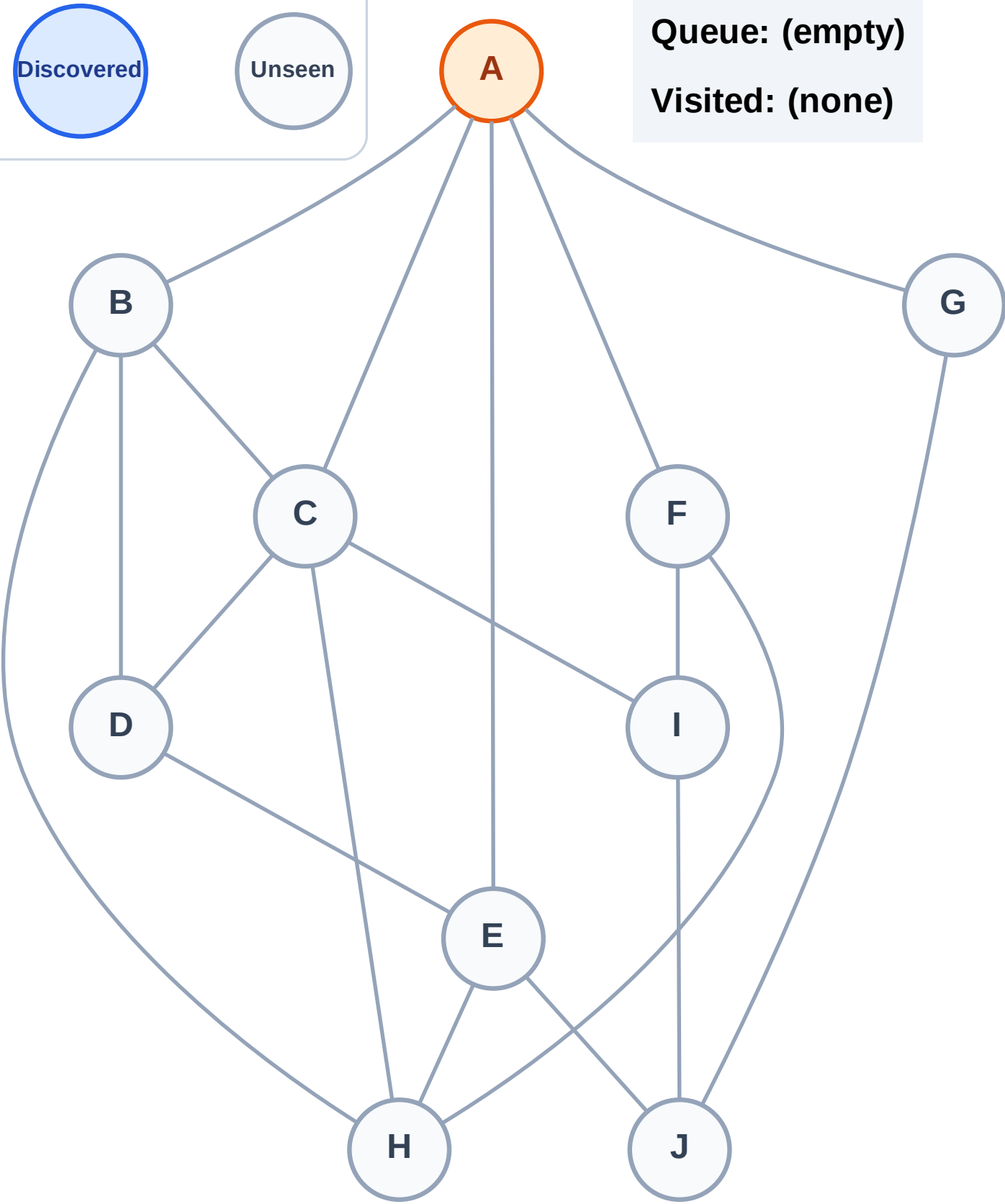
Complete

Processing

Discovered

Unseen

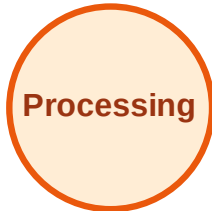
Queue: (empty)
Visited: (none)



BFS Traversal

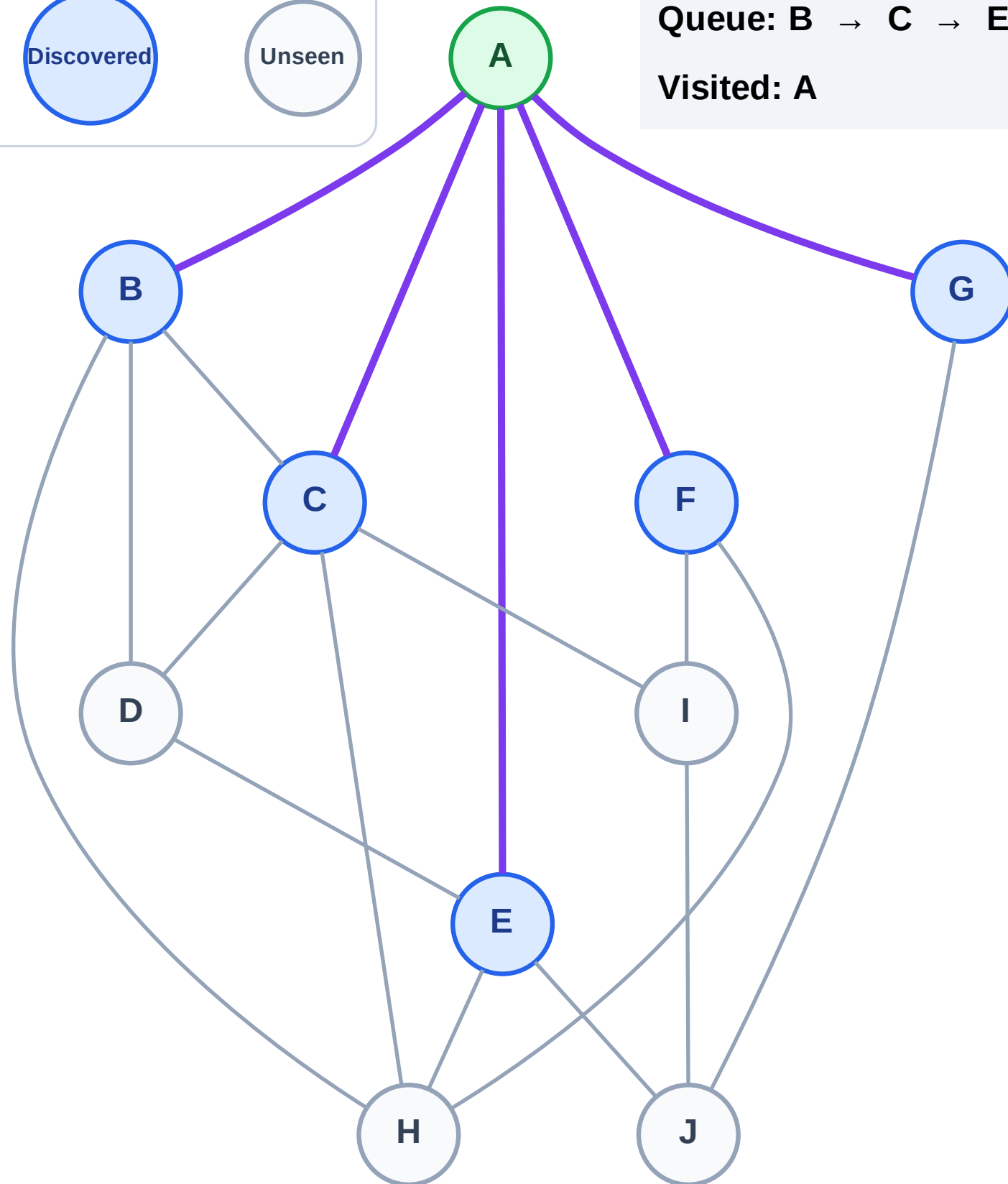
Step 3: Discover B, C, E, F, G from A

Legend



Queue: B → C → E → F → G

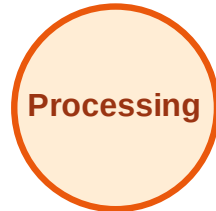
Visited: A



BFS Traversal

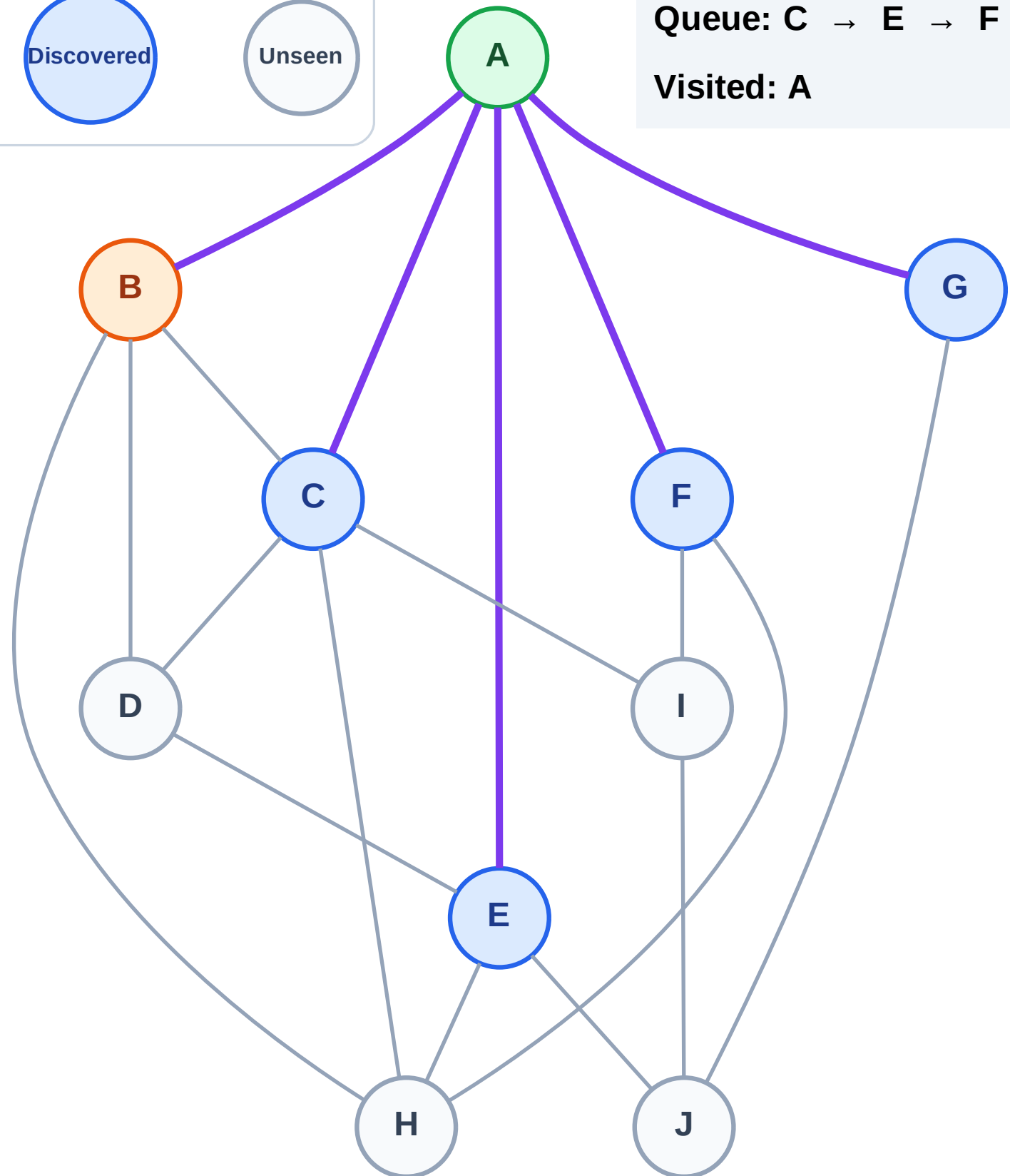
Step 4: Process node B

Legend



Queue: C → E → F → G

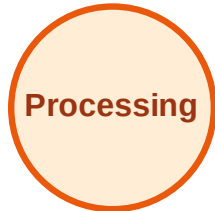
Visited: A



BFS Traversal

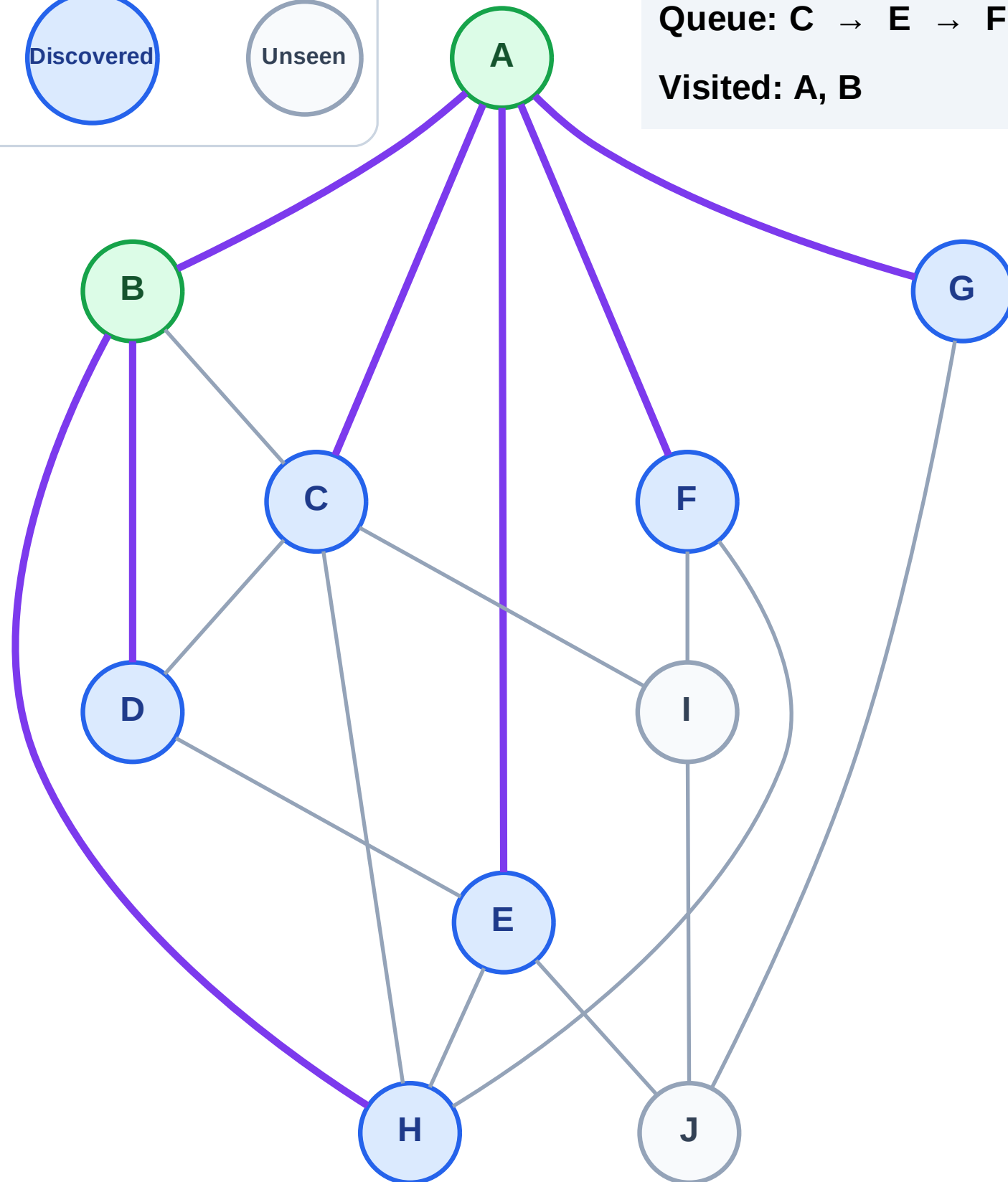
Step 5: Discover D, H from B

Legend



Queue: C → E → F → G → D → H

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

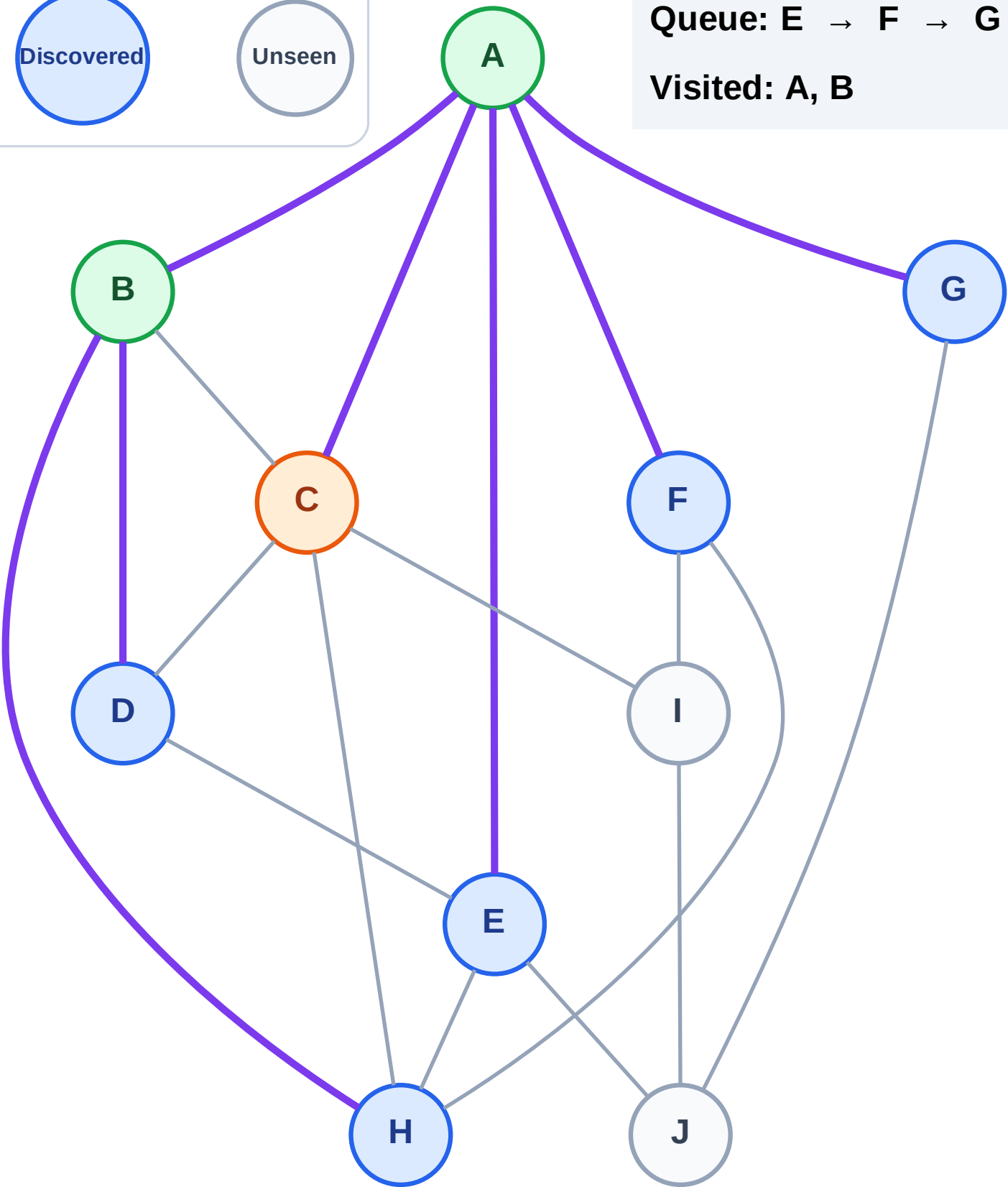
Processing

Discovered

Unseen

Queue: E → F → G → D → H

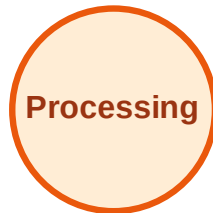
Visited: A, B



BFS Traversal

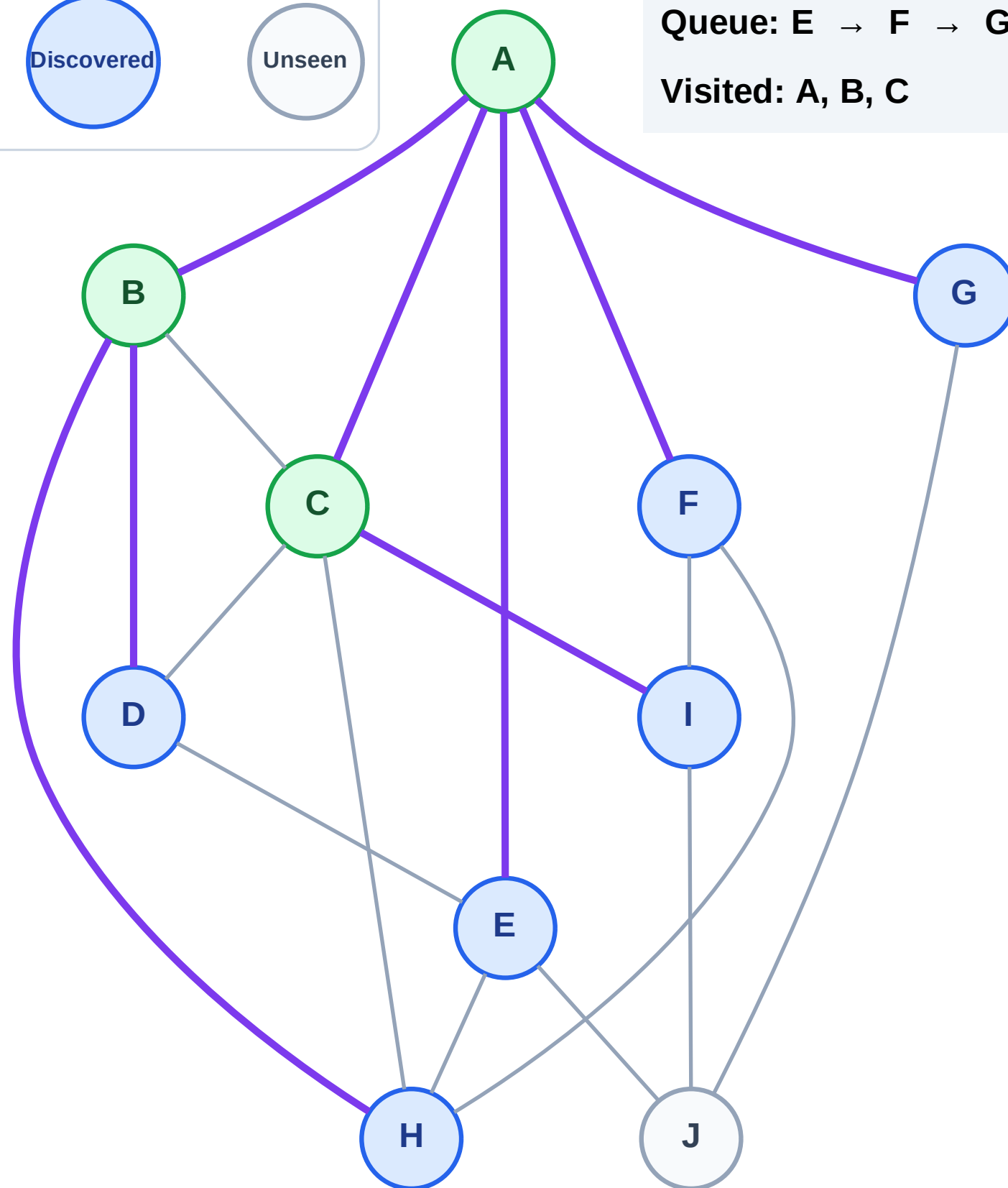
Step 7: Discover I from C

Legend



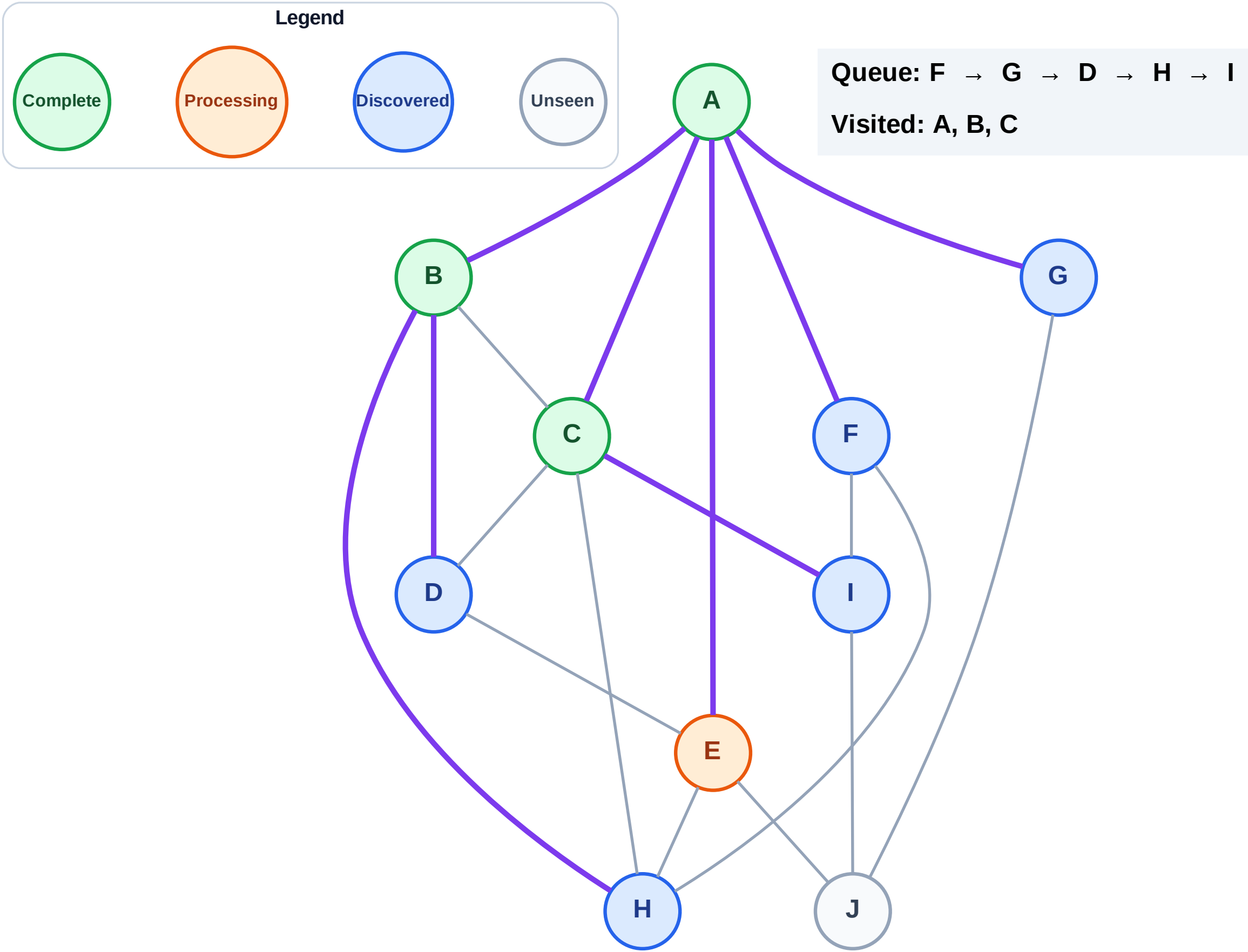
Queue: E → F → G → D → H → I

Visited: A, B, C



BFS Traversal

Step 8: Process node E

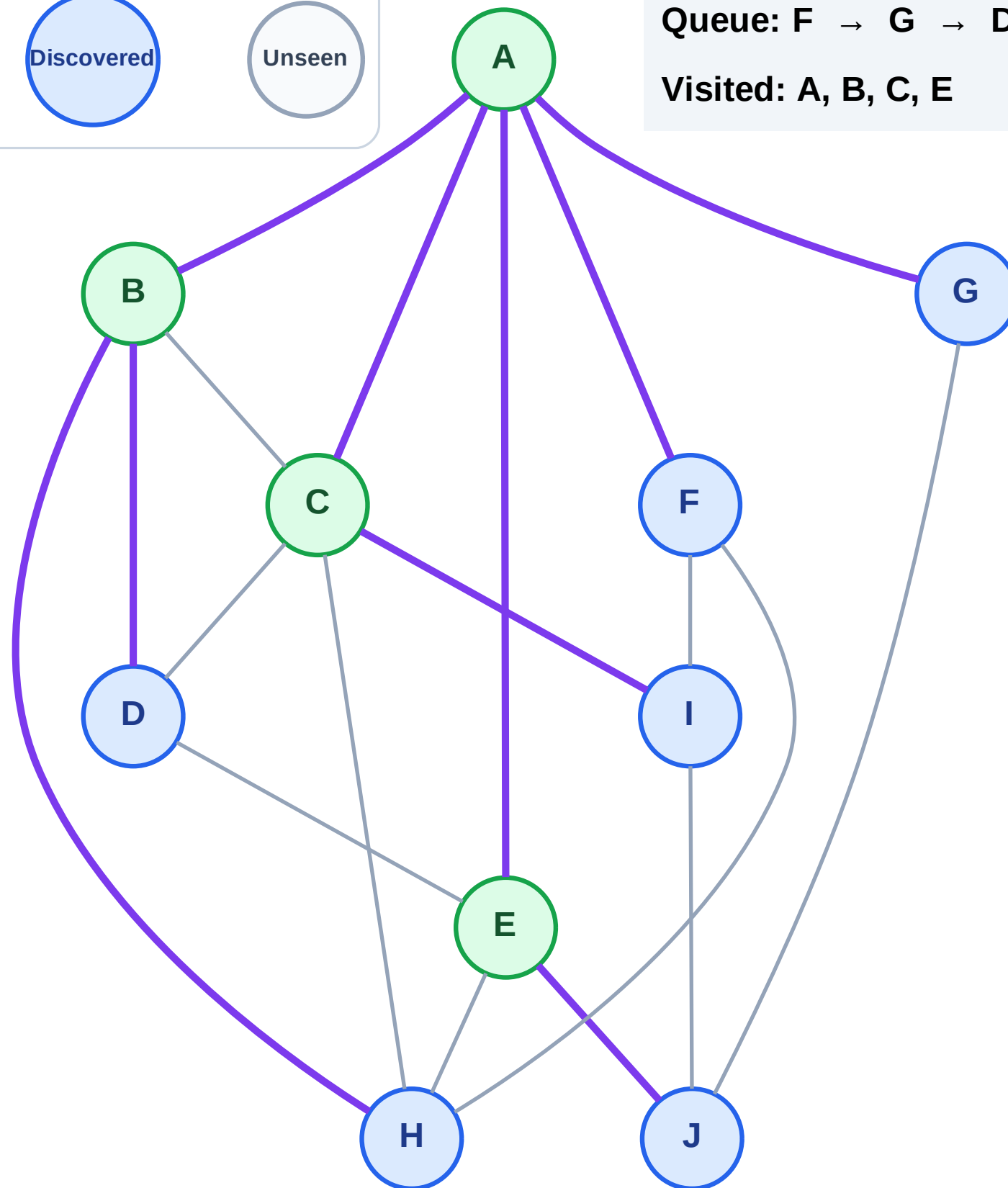


Step 9: Discover J from E

Legend

Unseen

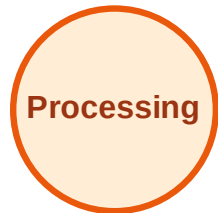
Visited: A, B, C, E



BFS Traversal

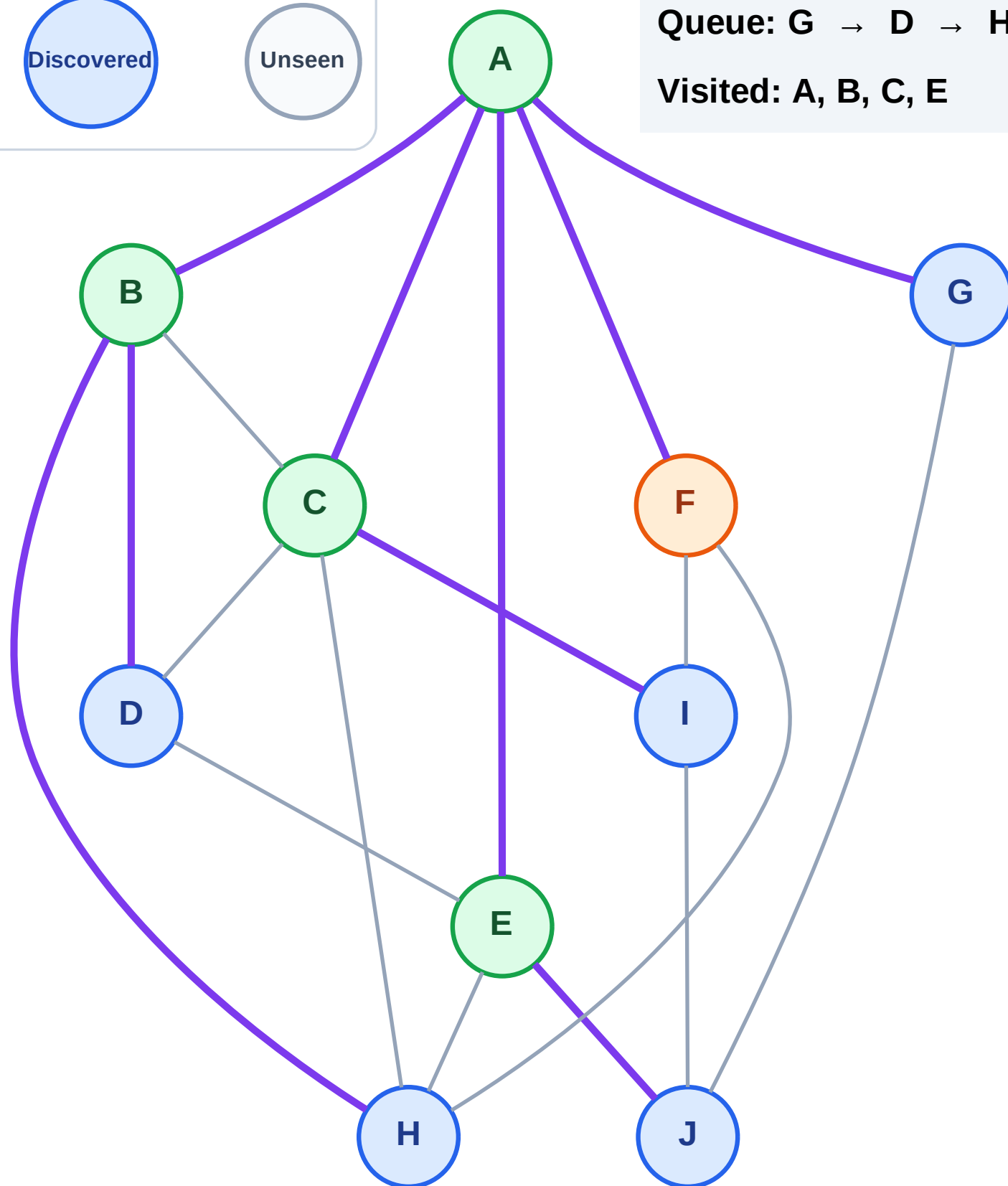
Step 10: Process node F

Legend



Queue: G → D → H → I → J

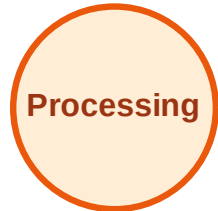
Visited: A, B, C, E



BFS Traversal

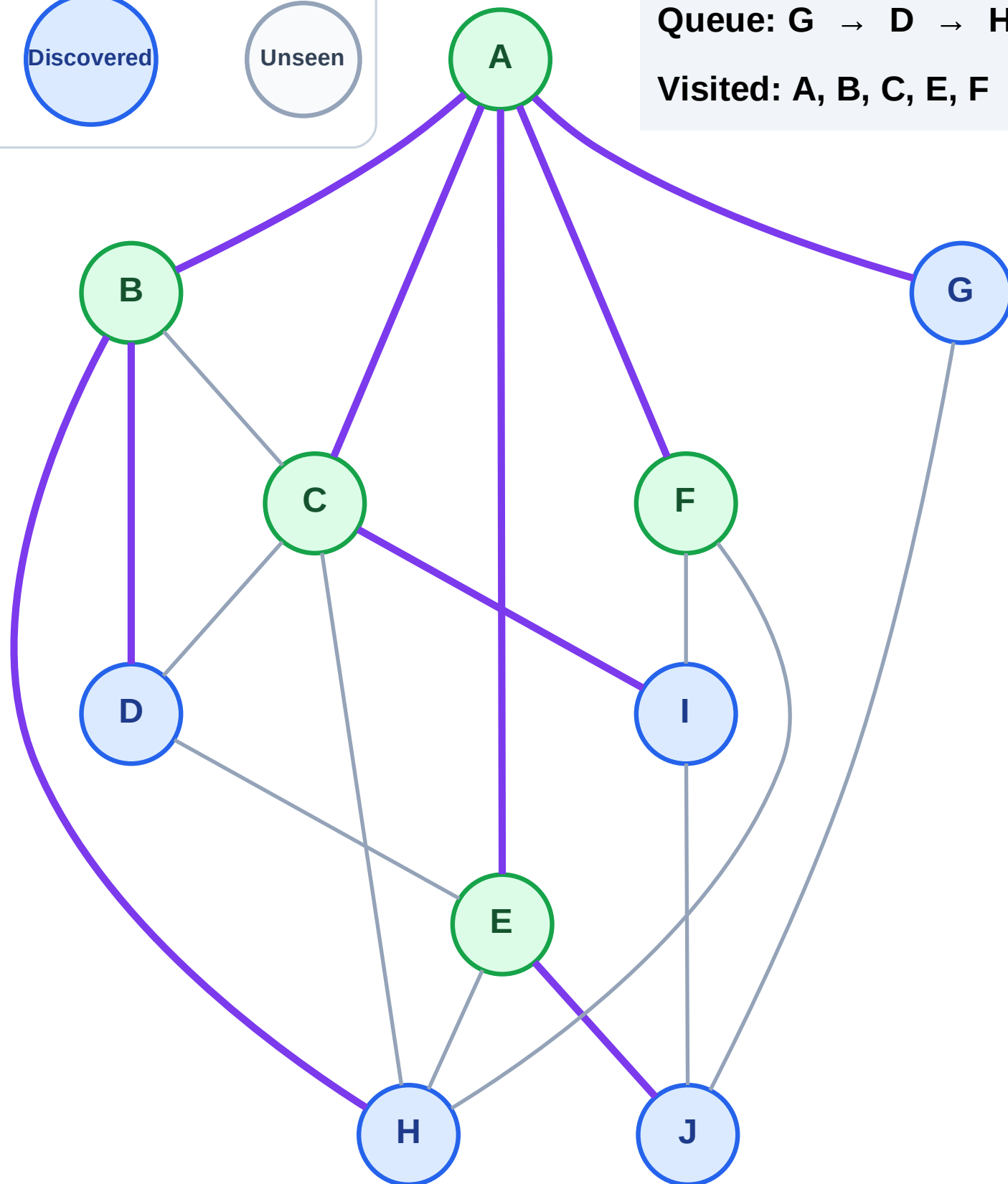
Step 11: No new neighbors from F

Legend



Queue: G → D → H → I → J

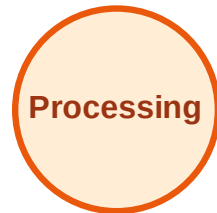
Visited: A, B, C, E, F



BFS Traversal

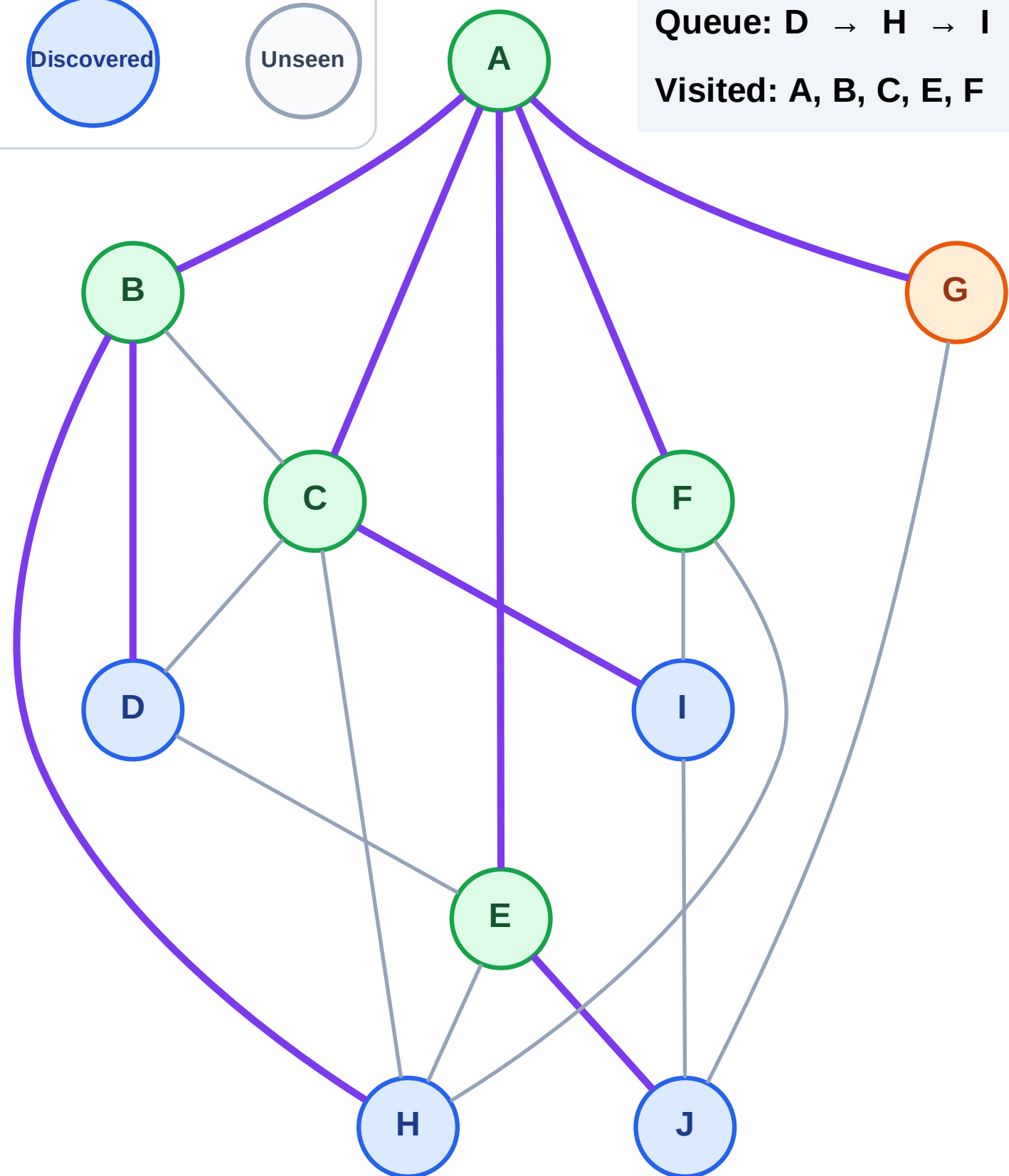
Step 12: Process node G

Legend



Queue: D → H → I → J

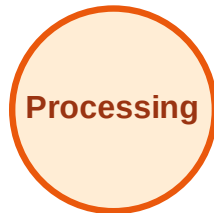
Visited: A, B, C, E, F



BFS Traversal

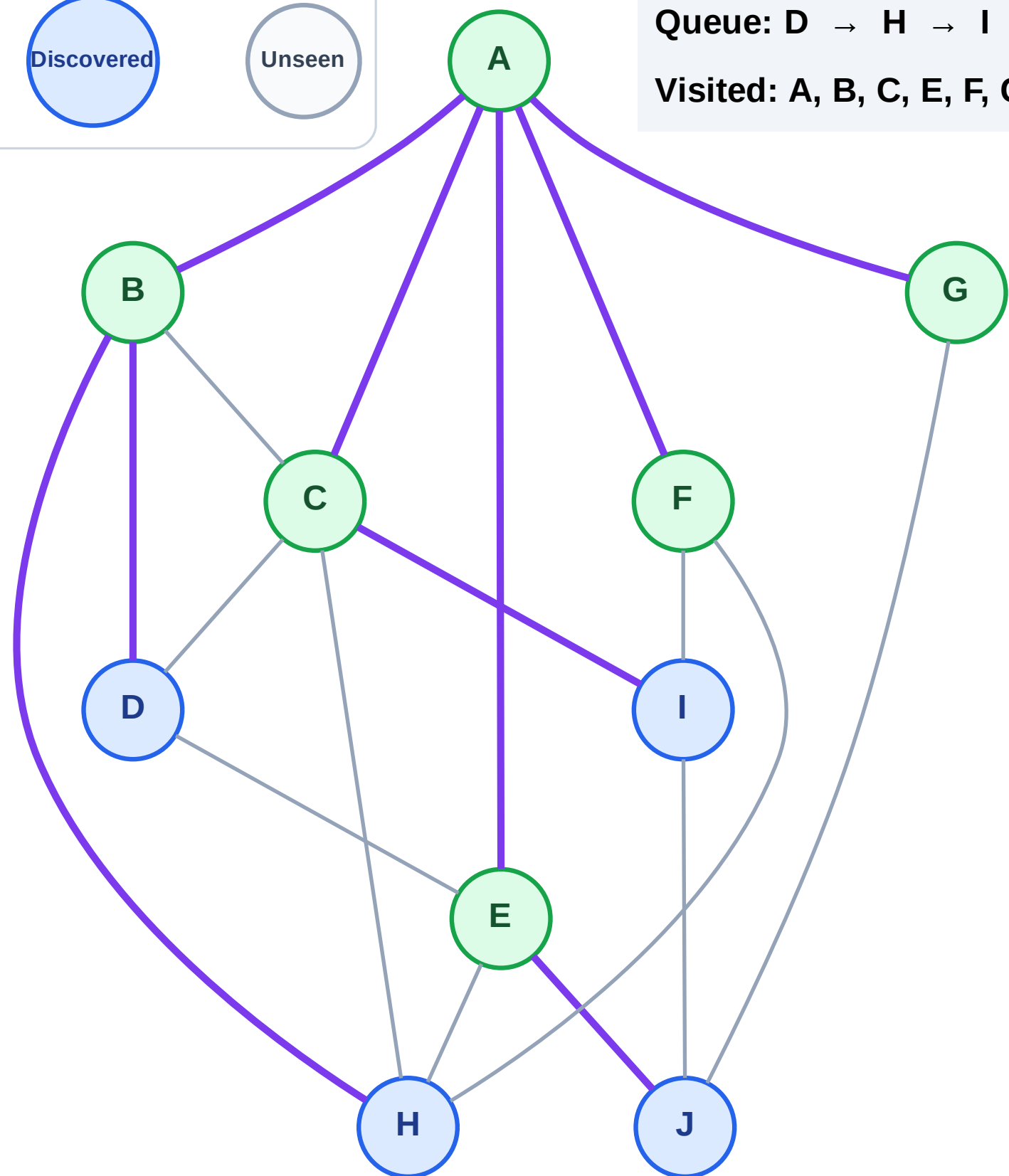
Step 13: No new neighbors from G

Legend



Queue: D → H → I → J

Visited: A, B, C, E, F, G



BFS Traversal

Step 14: Process node D

Legend

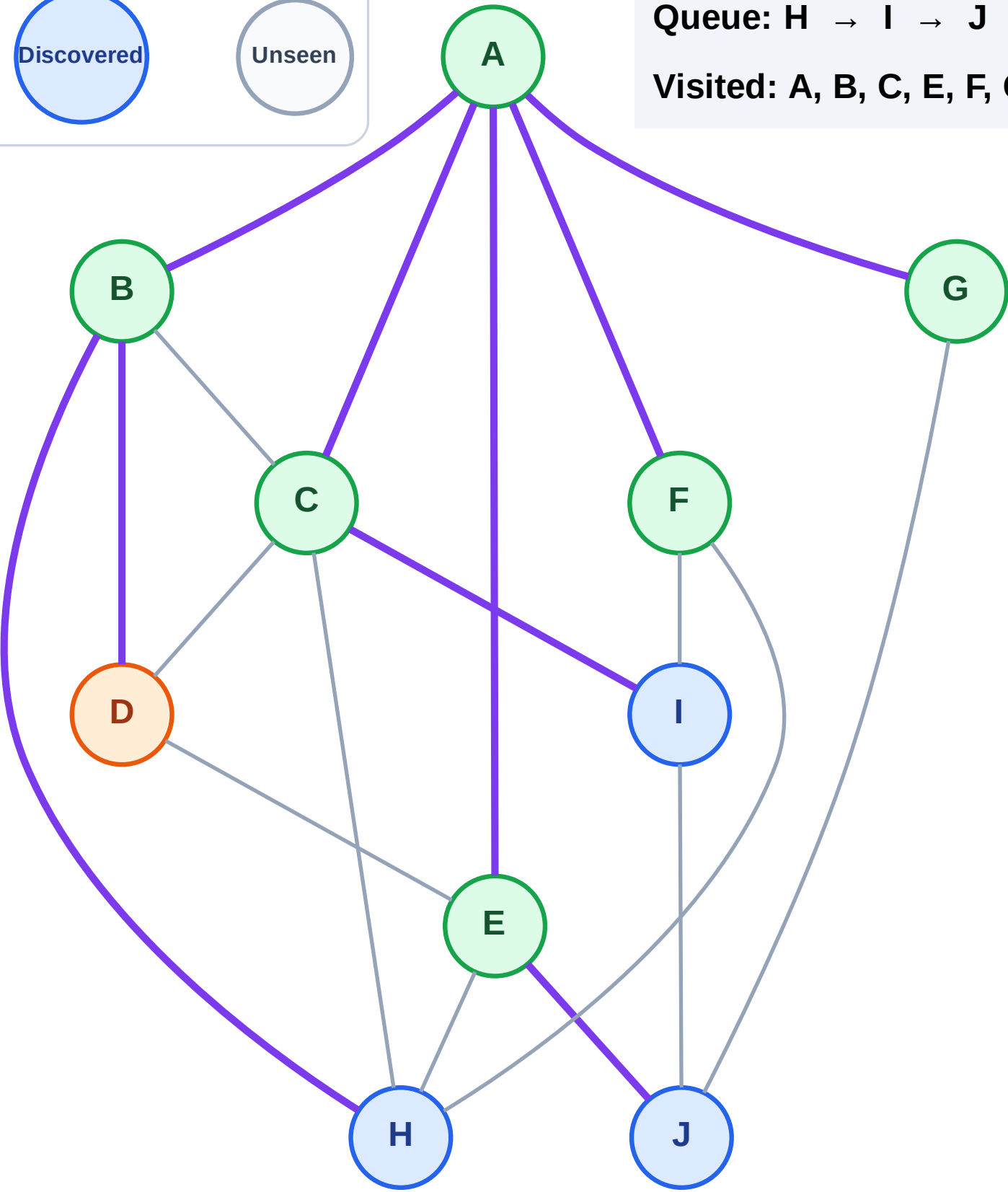
Complete

Processing

Discovered

Unseen

Queue: H → I → J
Visited: A, B, C, E, F, G



BFS Traversal

Step 15: No new neighbors from D

Legend

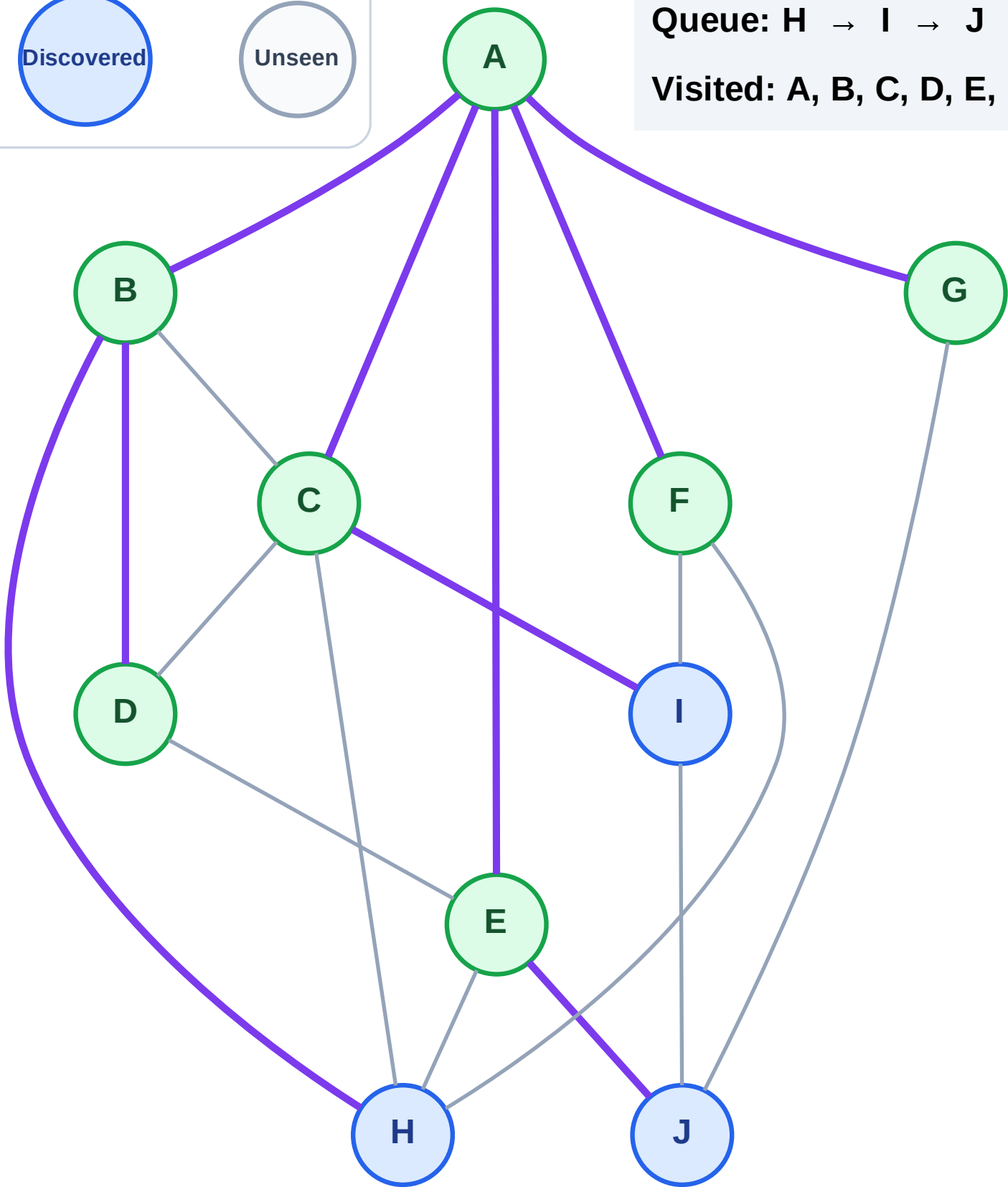
Complete

Processing

Discovered

Unseen

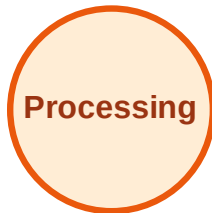
Queue: H → I → J
Visited: A, B, C, D, E, F, G



BFS Traversal

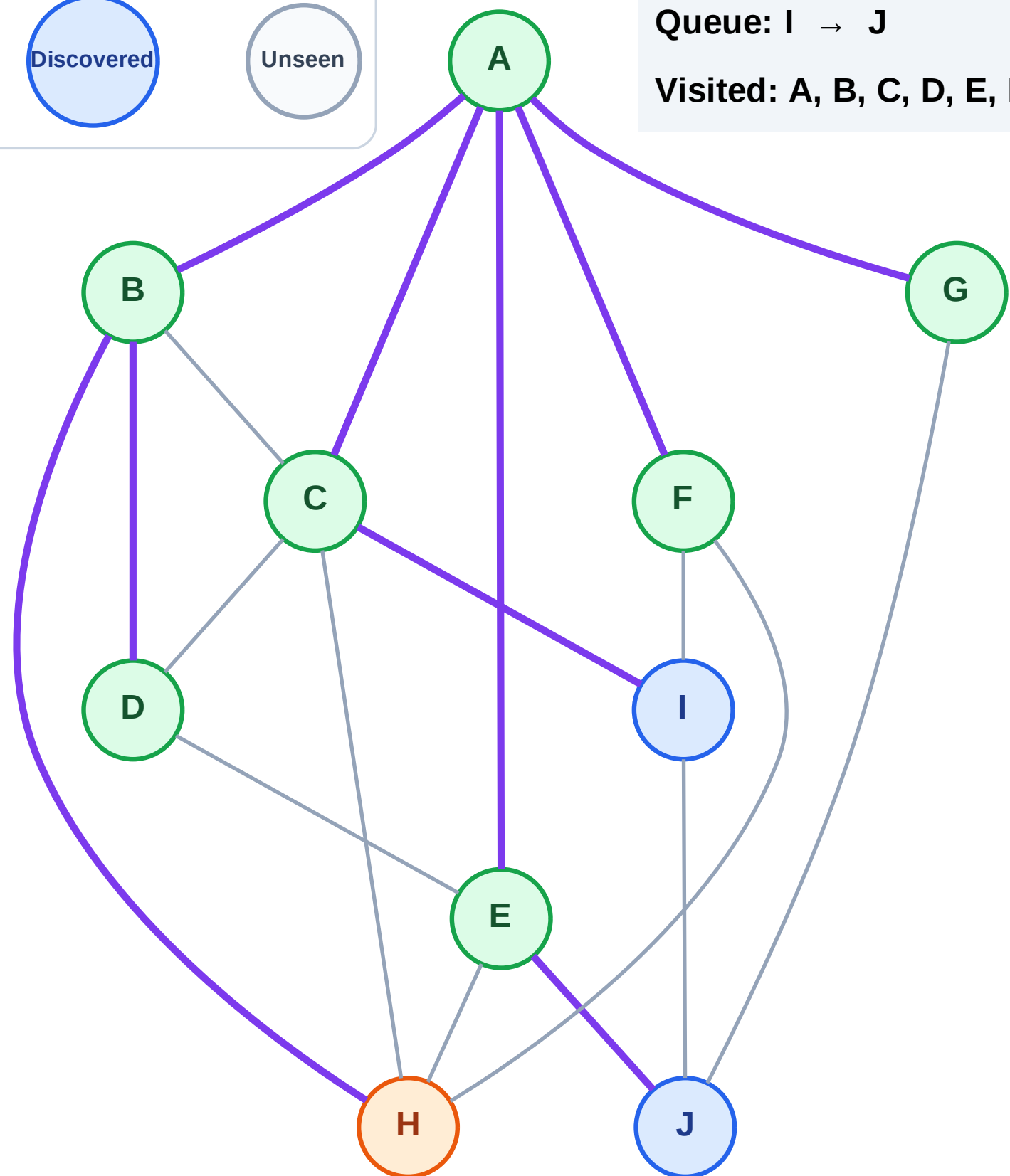
Step 16: Process node H

Legend



Queue: I → J

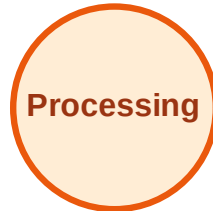
Visited: A, B, C, D, E, F, G



BFS Traversal

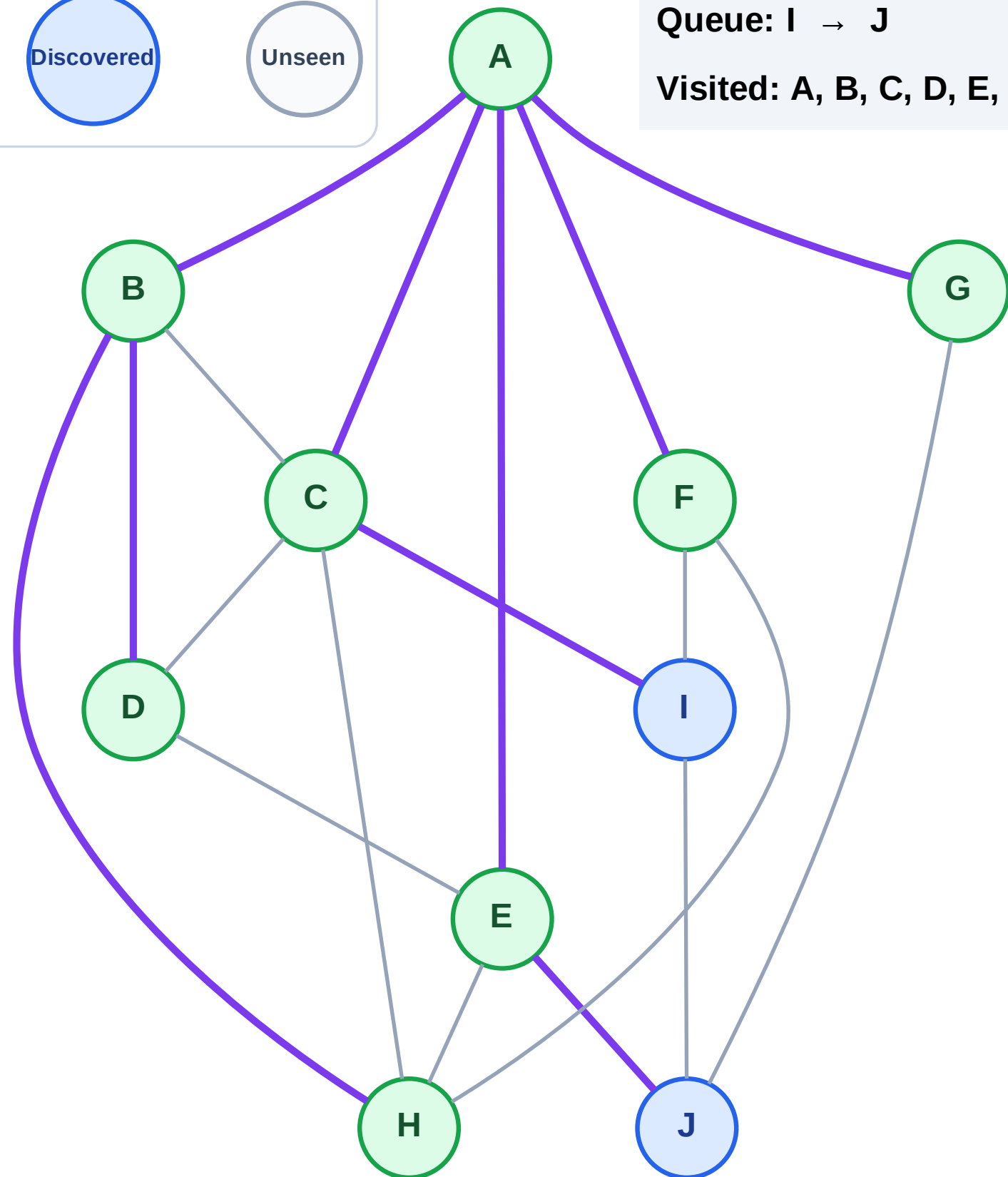
Step 17: No new neighbors from H

Legend



Queue: I → J

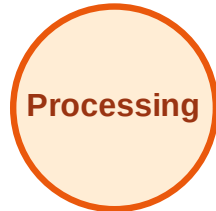
Visited: A, B, C, D, E, F, G, H



BFS Traversal

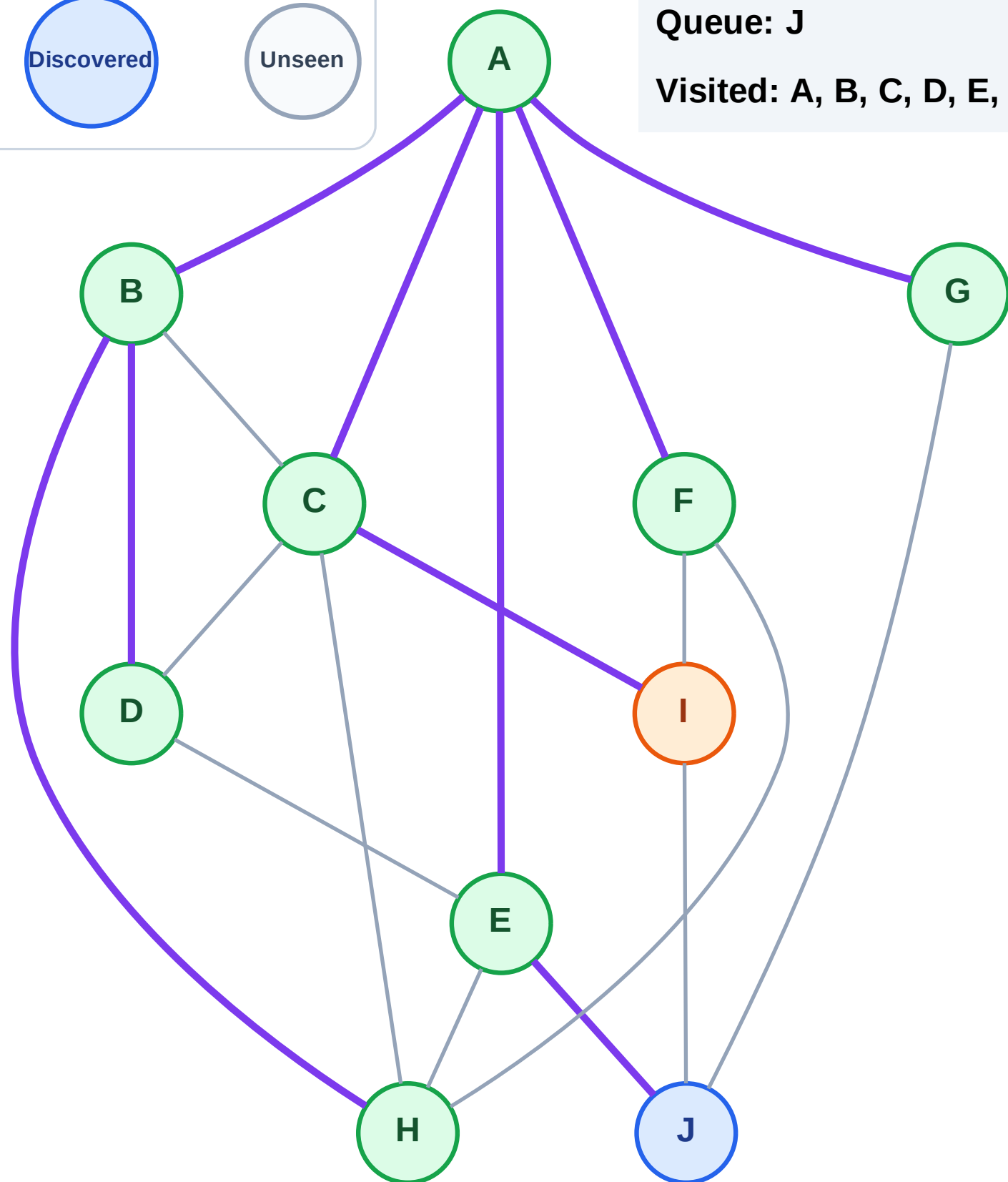
Step 18: Process node I

Legend



Queue: J

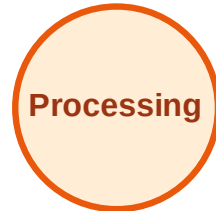
Visited: A, B, C, D, E, F, G, H



BFS Traversal

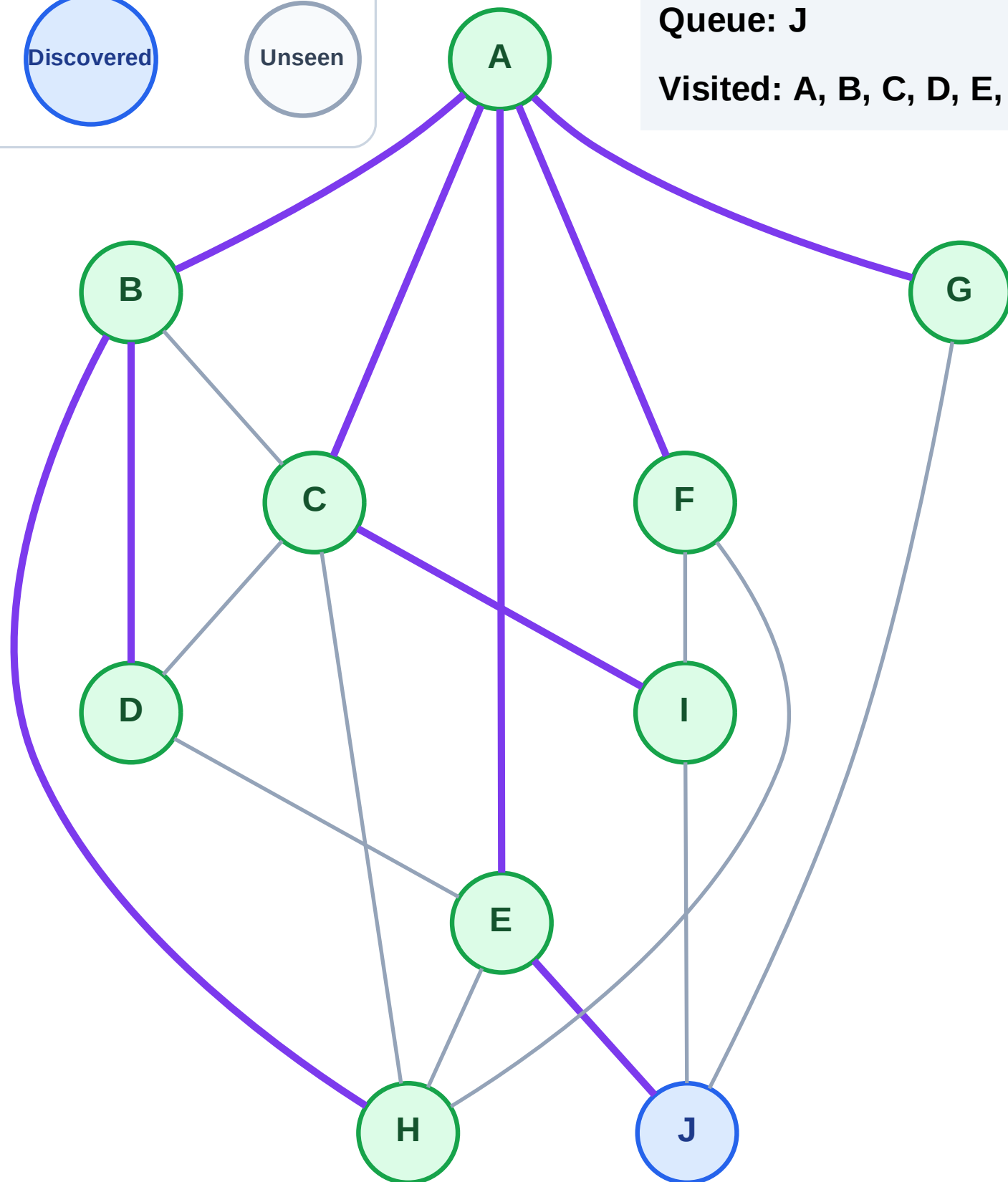
Step 19: No new neighbors from I

Legend



Queue: J

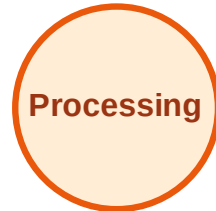
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

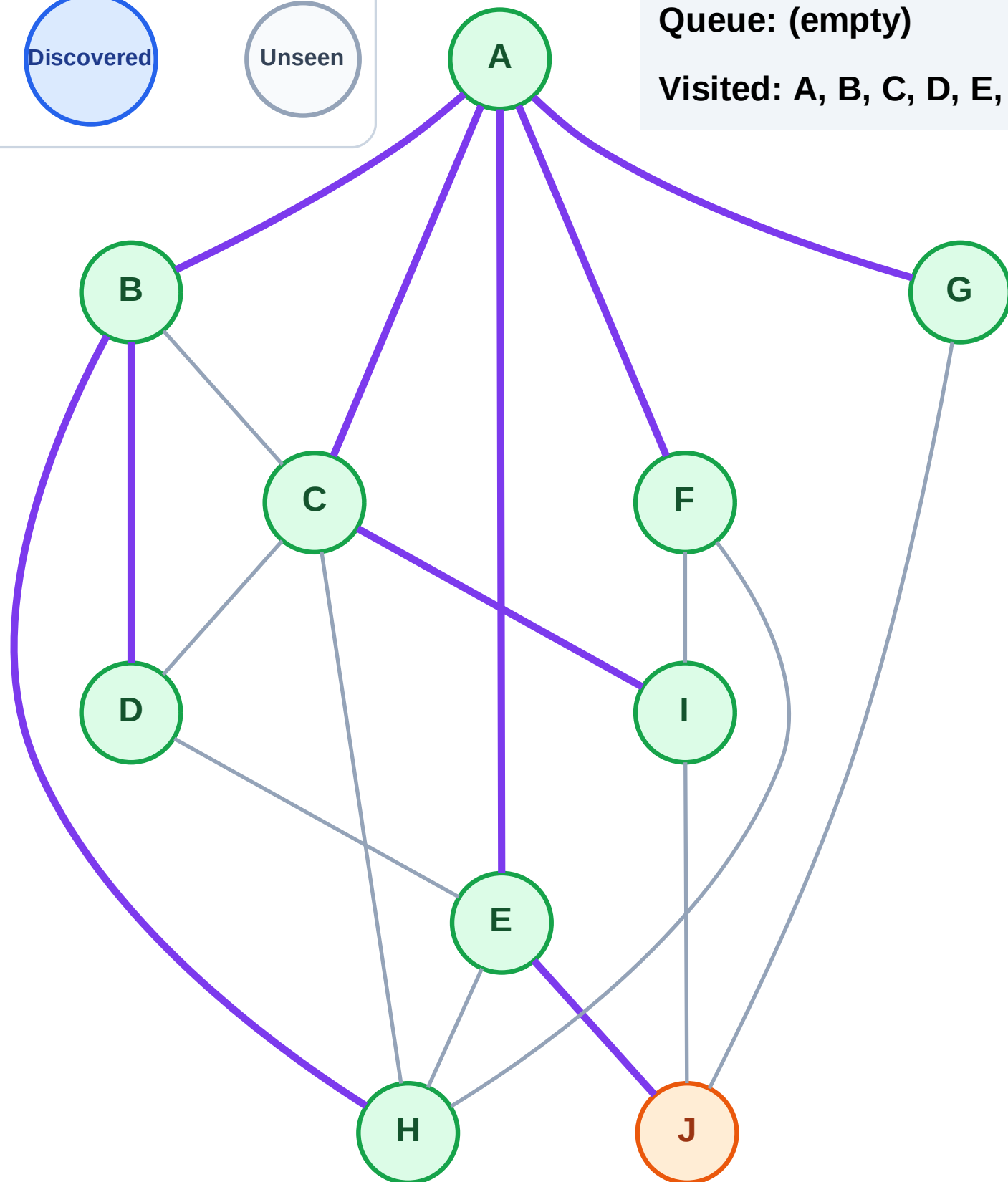
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

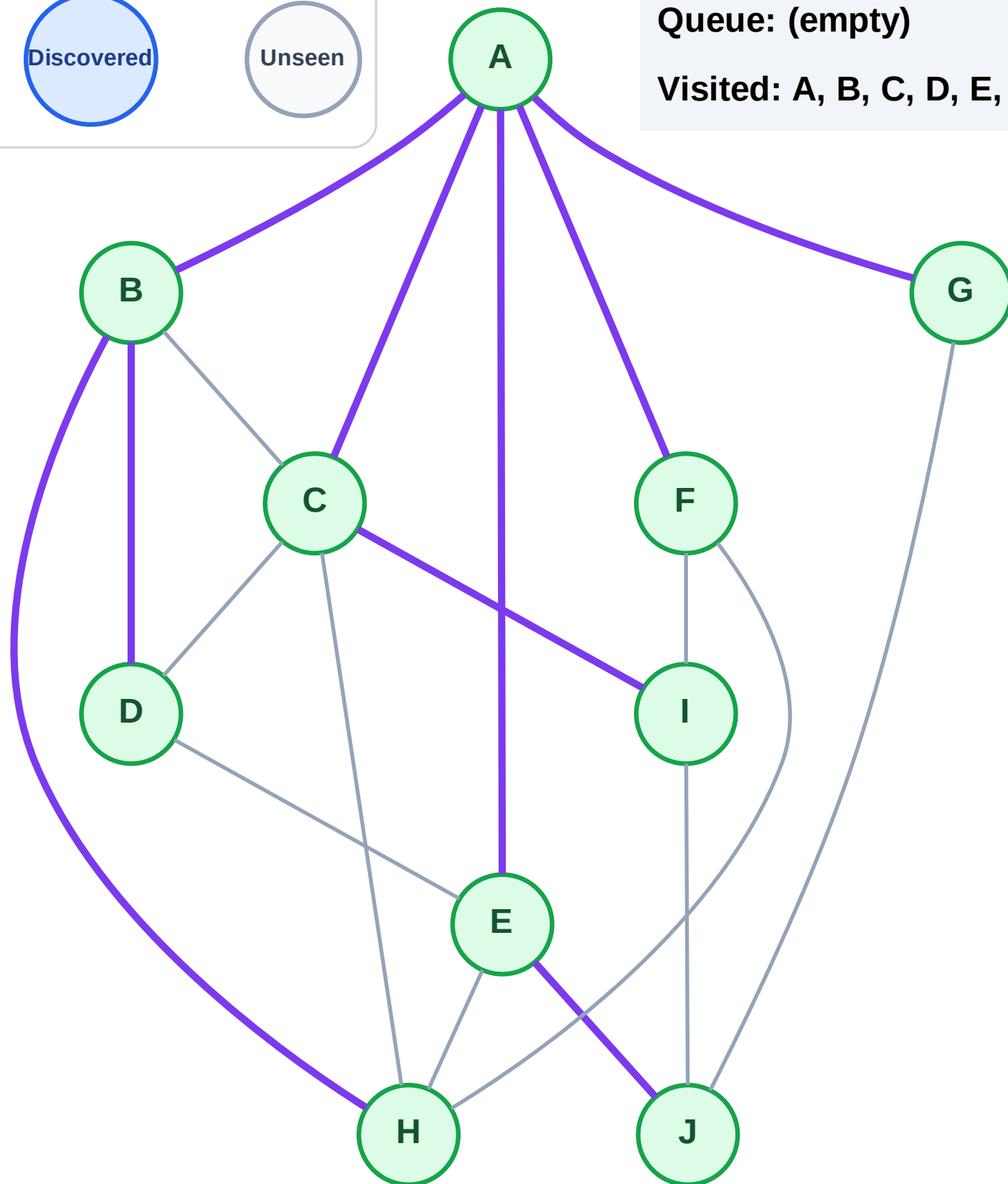
Step 21: No new neighbors from J

Legend



Queue: (empty)

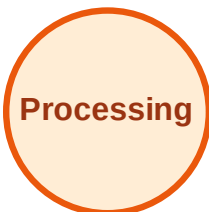
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

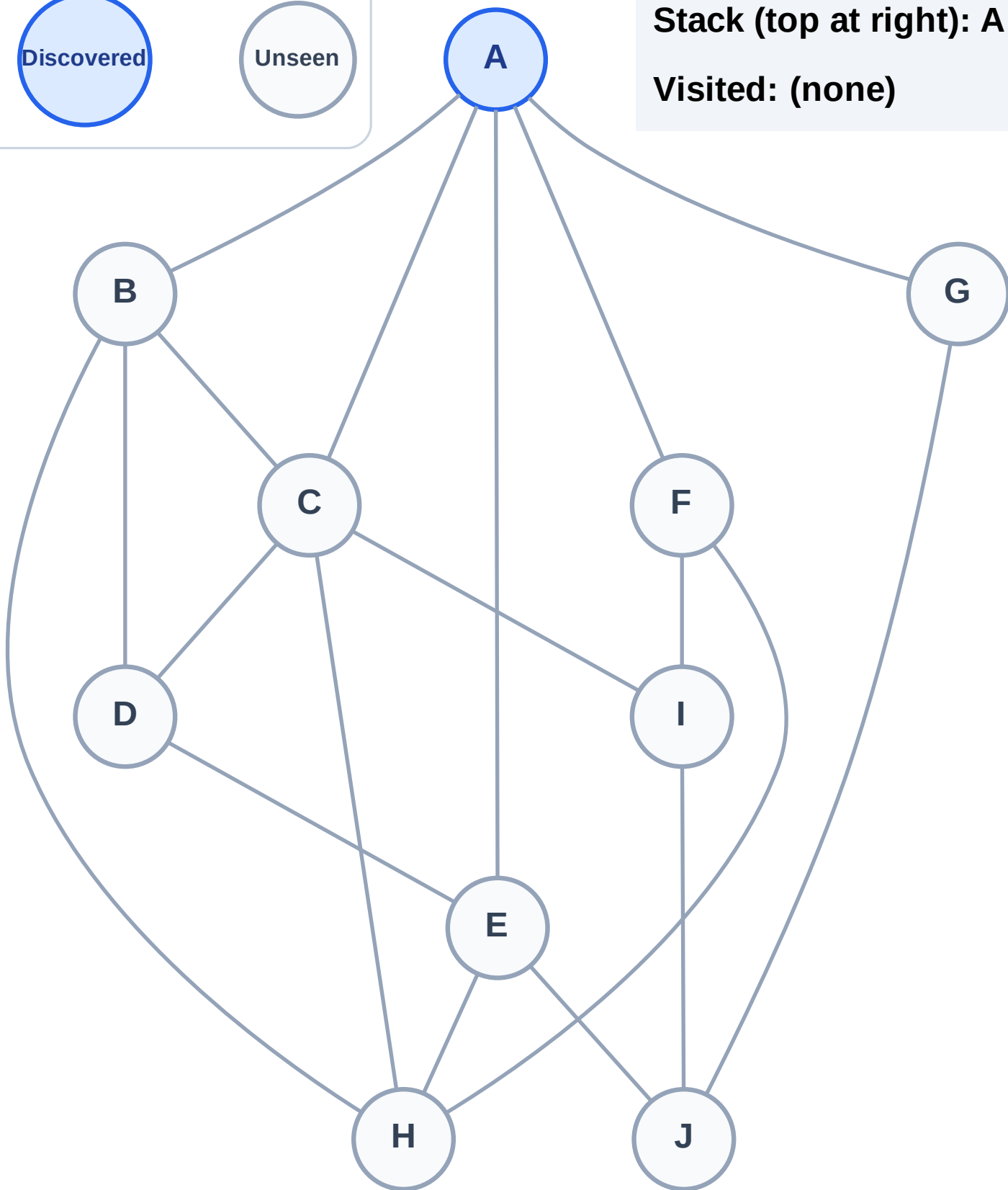
Step 1: Start at A

Legend



Stack (top at right): A

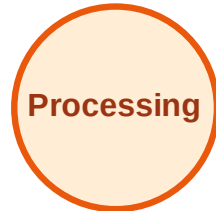
Visited: (none)



DFS Traversal

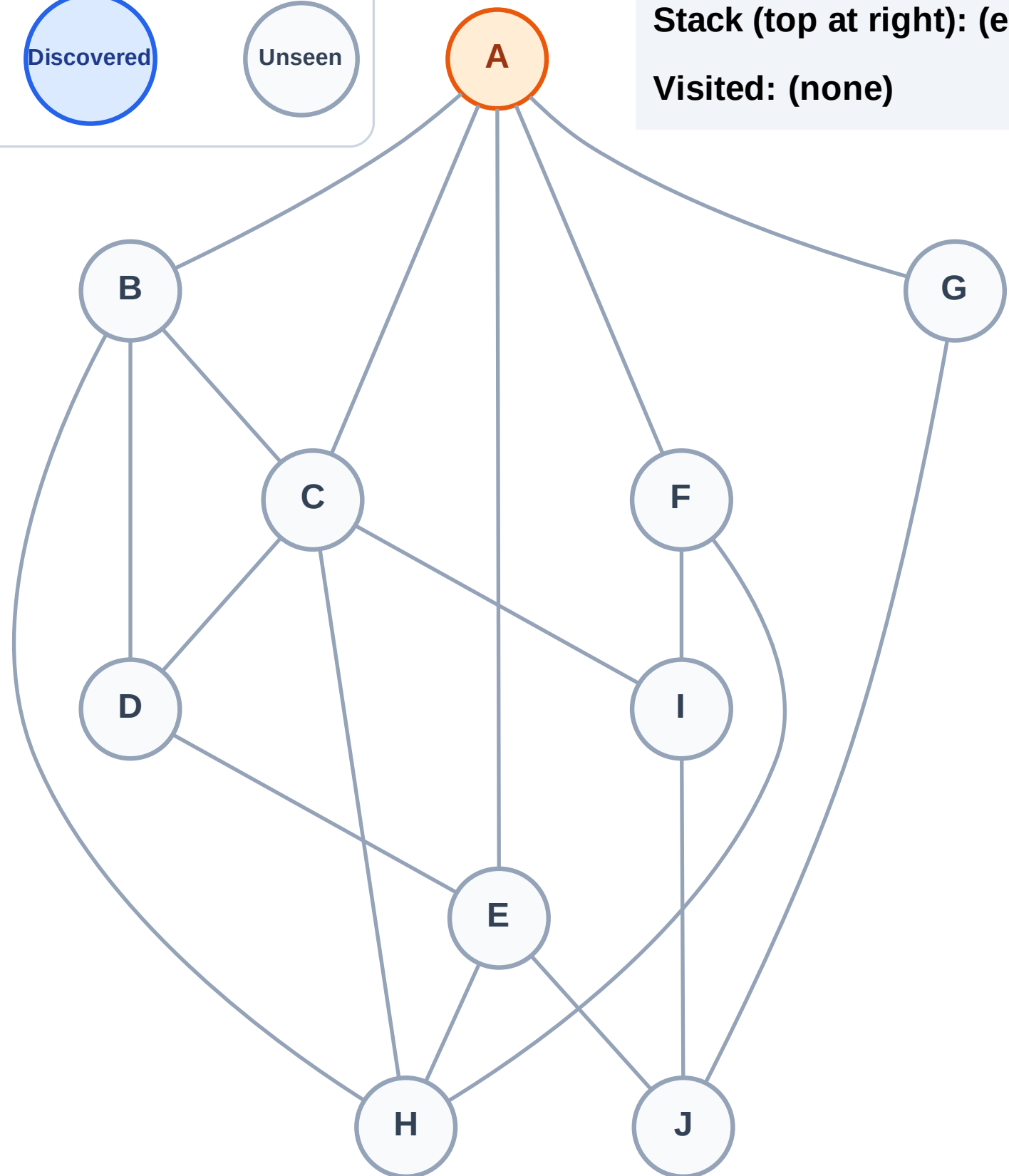
Step 2: Process node A

Legend



Stack (top at right): (empty)

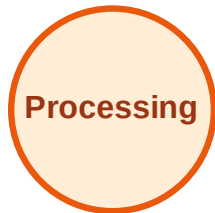
Visited: (none)



DFS Traversal

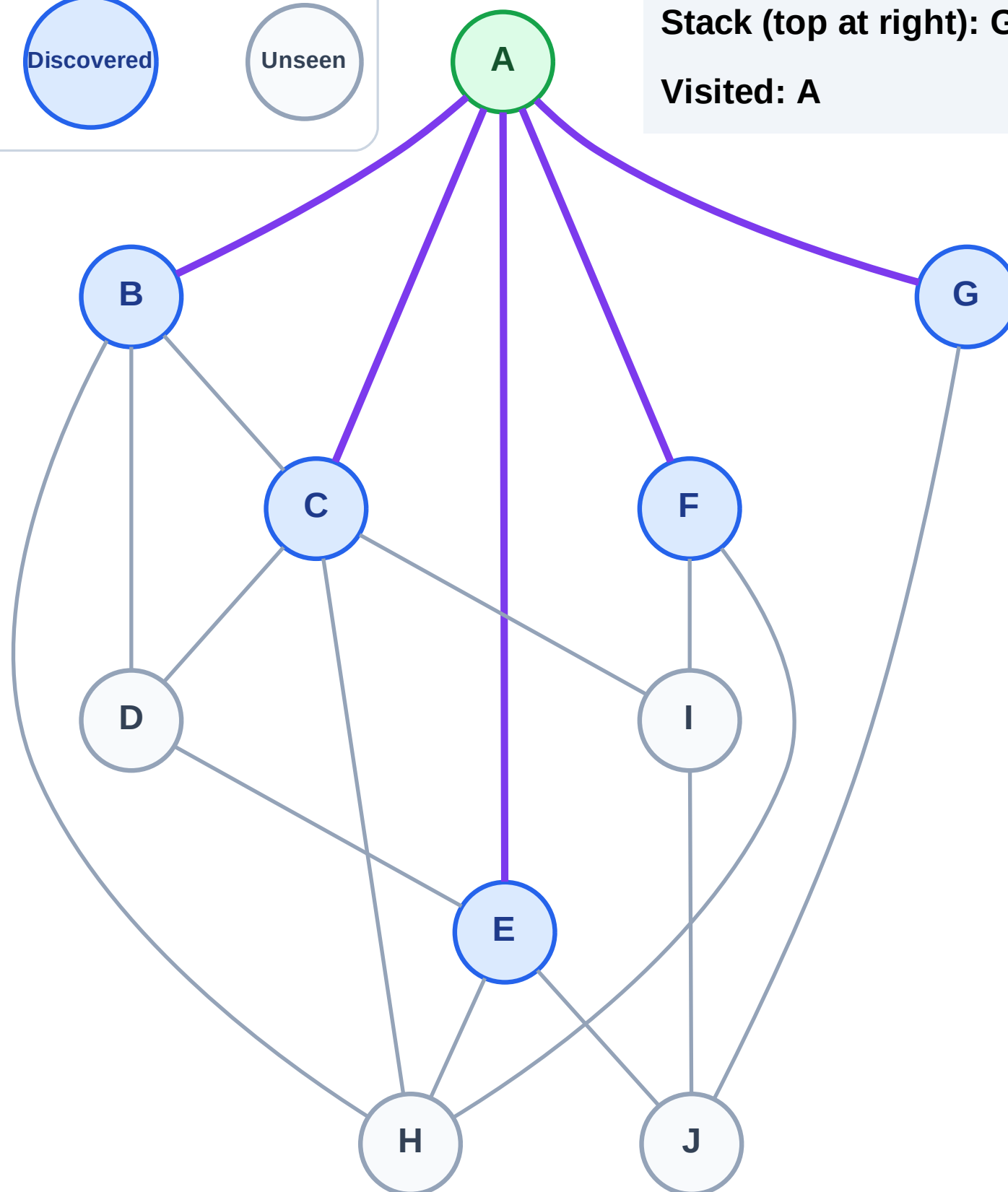
Step 3: Discover G, F, E, C, B from A

Legend



Stack (top at right): G | F | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

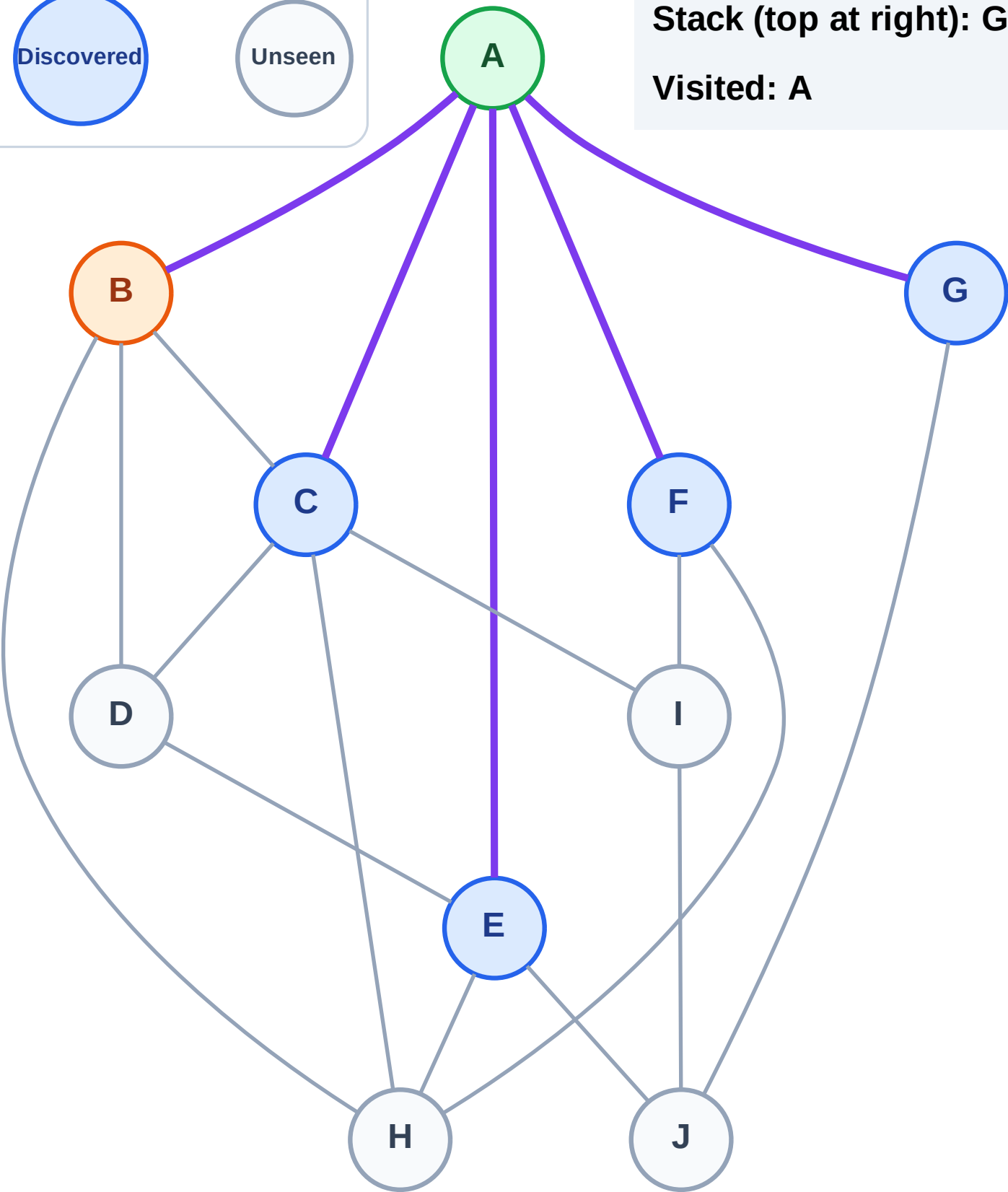
Complete

Processing

Discovered

Unseen

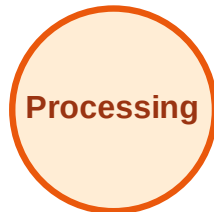
Stack (top at right): G | F | E | C
Visited: A



DFS Traversal

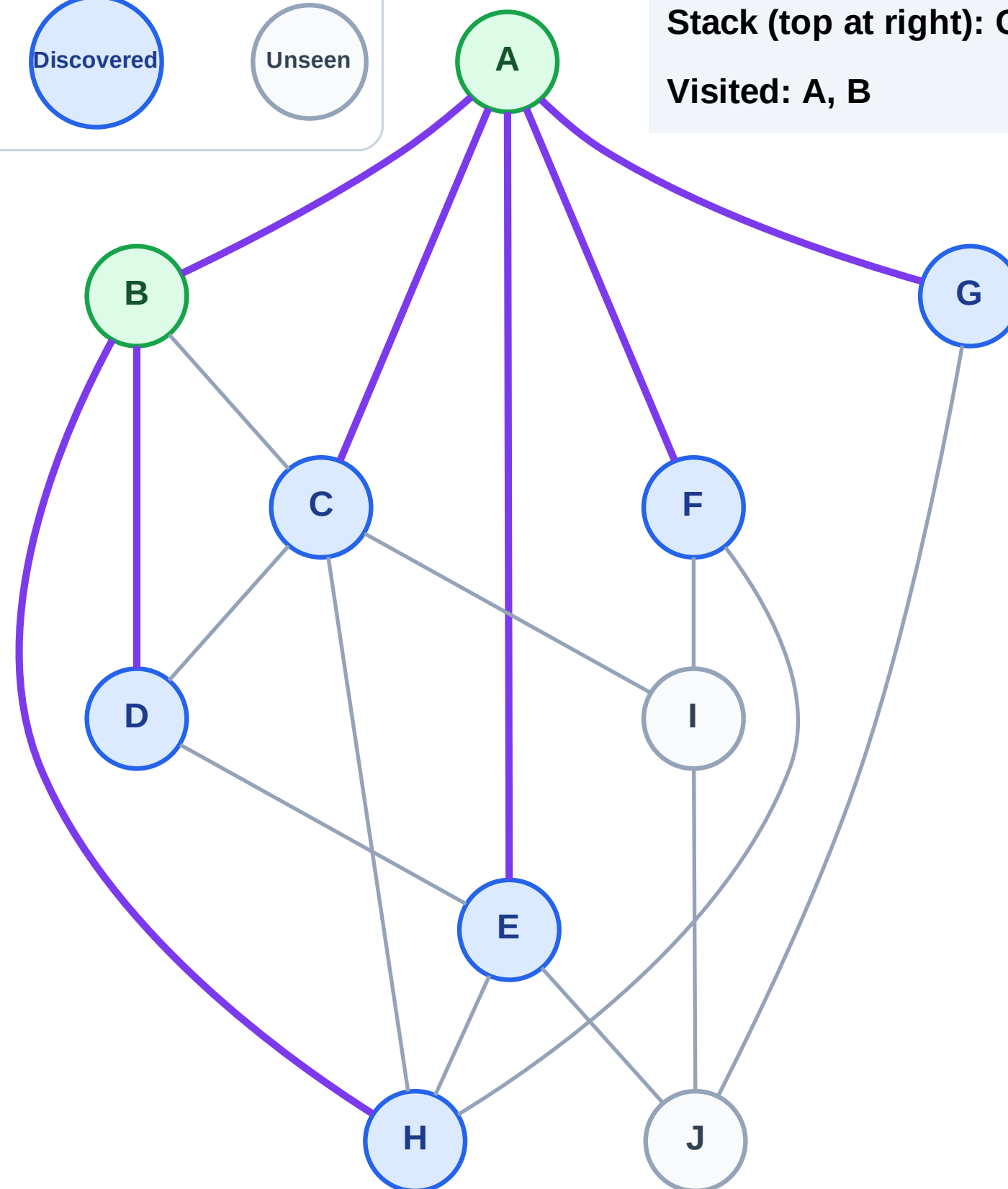
Step 5: Discover H, D from B

Legend



Stack (top at right): G | F | E | C | H | D

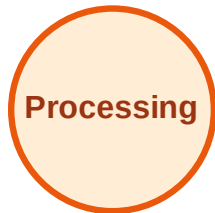
Visited: A, B



DFS Traversal

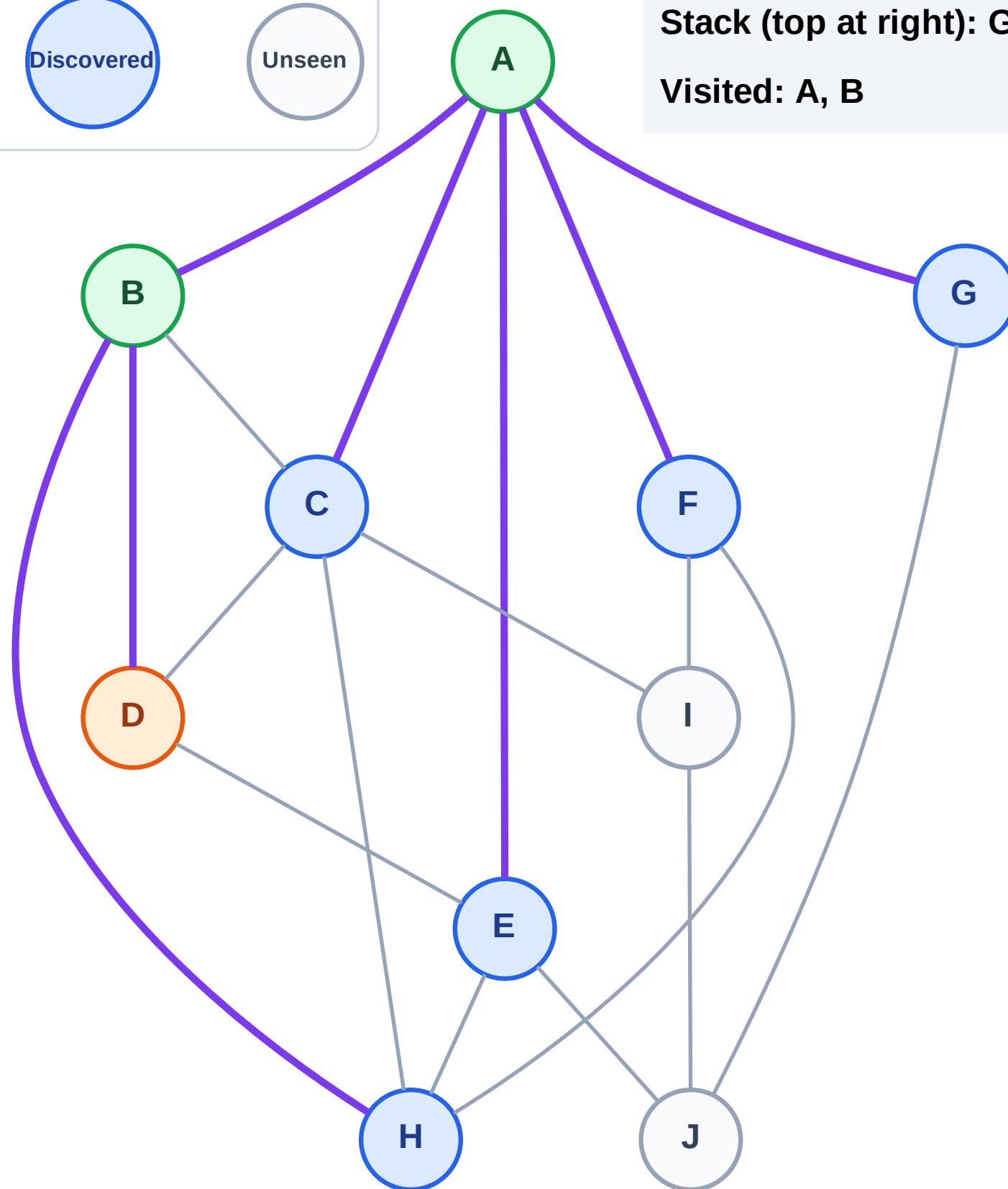
Step 6: Process node D

Legend



Stack (top at right): G | F | E | C | H

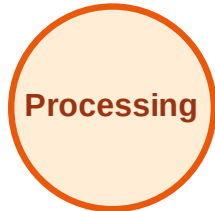
Visited: A, B



DFS Traversal

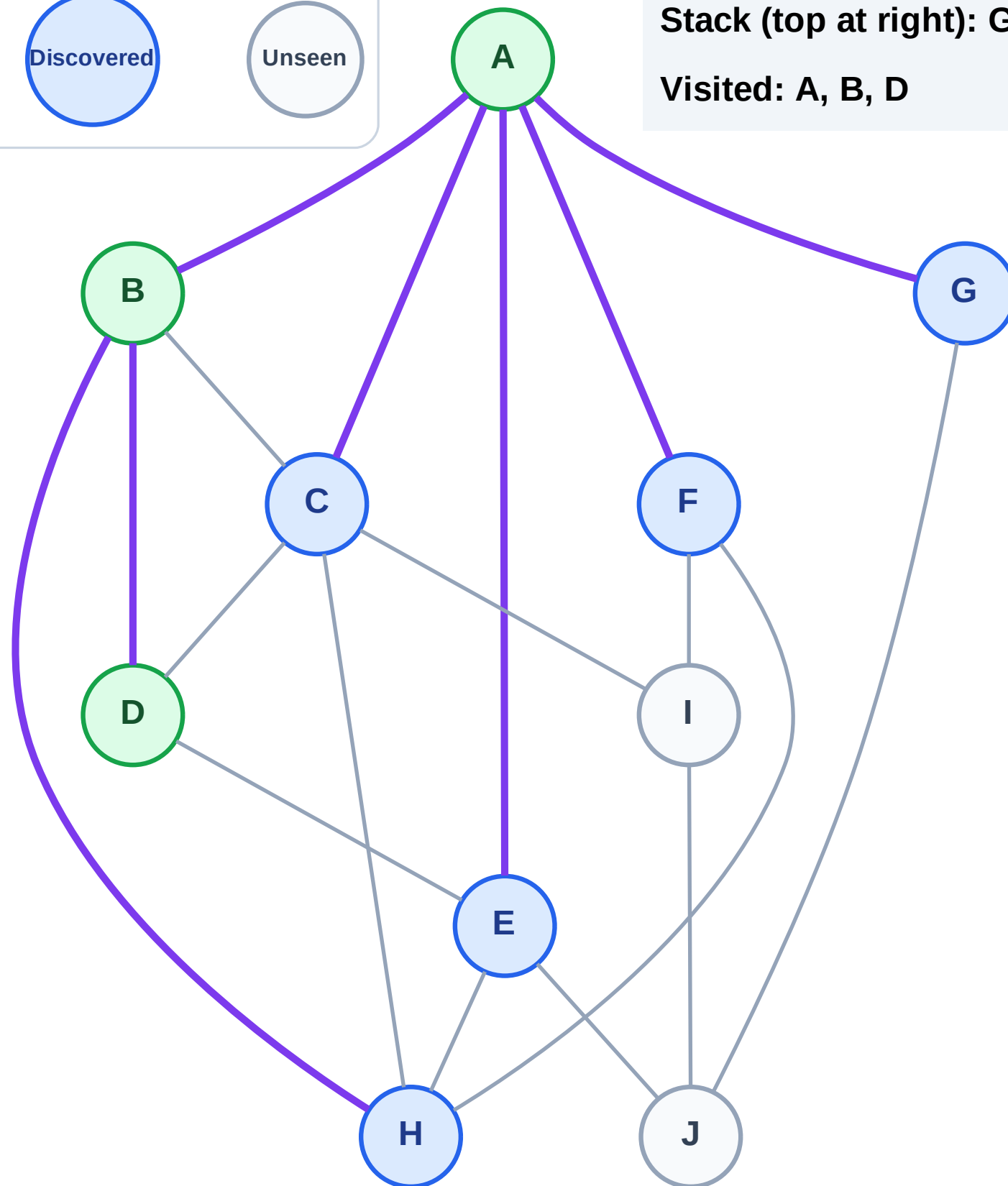
Step 7: No new neighbors from D

Legend



Stack (top at right): G | F | E | C | H

Visited: A, B, D



DFS Traversal

Step 8: Process node H

Legend

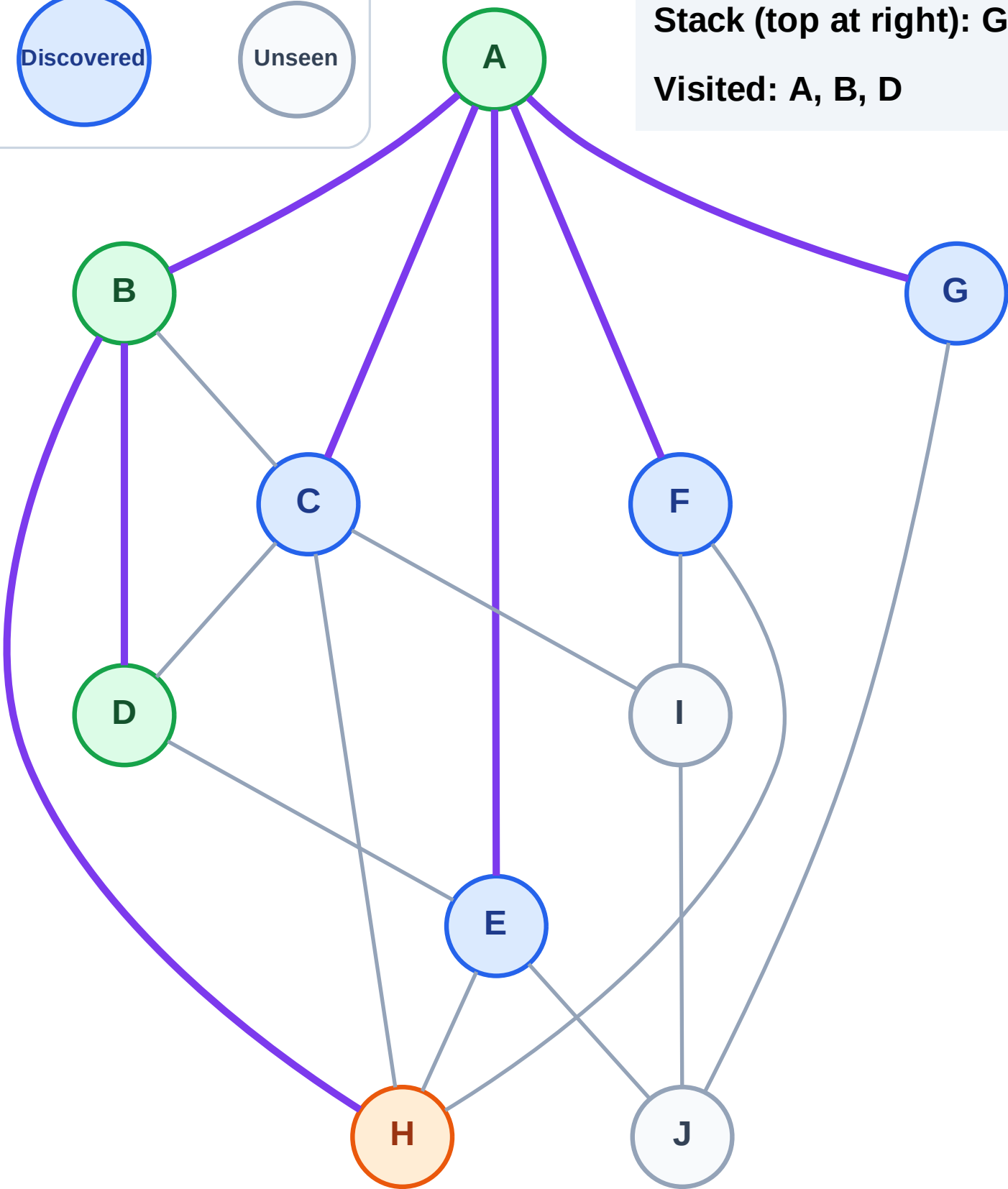
Complete

Processing

Discovered

Unseen

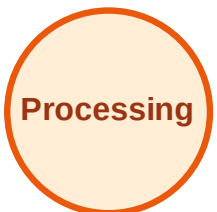
Stack (top at right): G | F | E | C
Visited: A, B, D



DFS Traversal

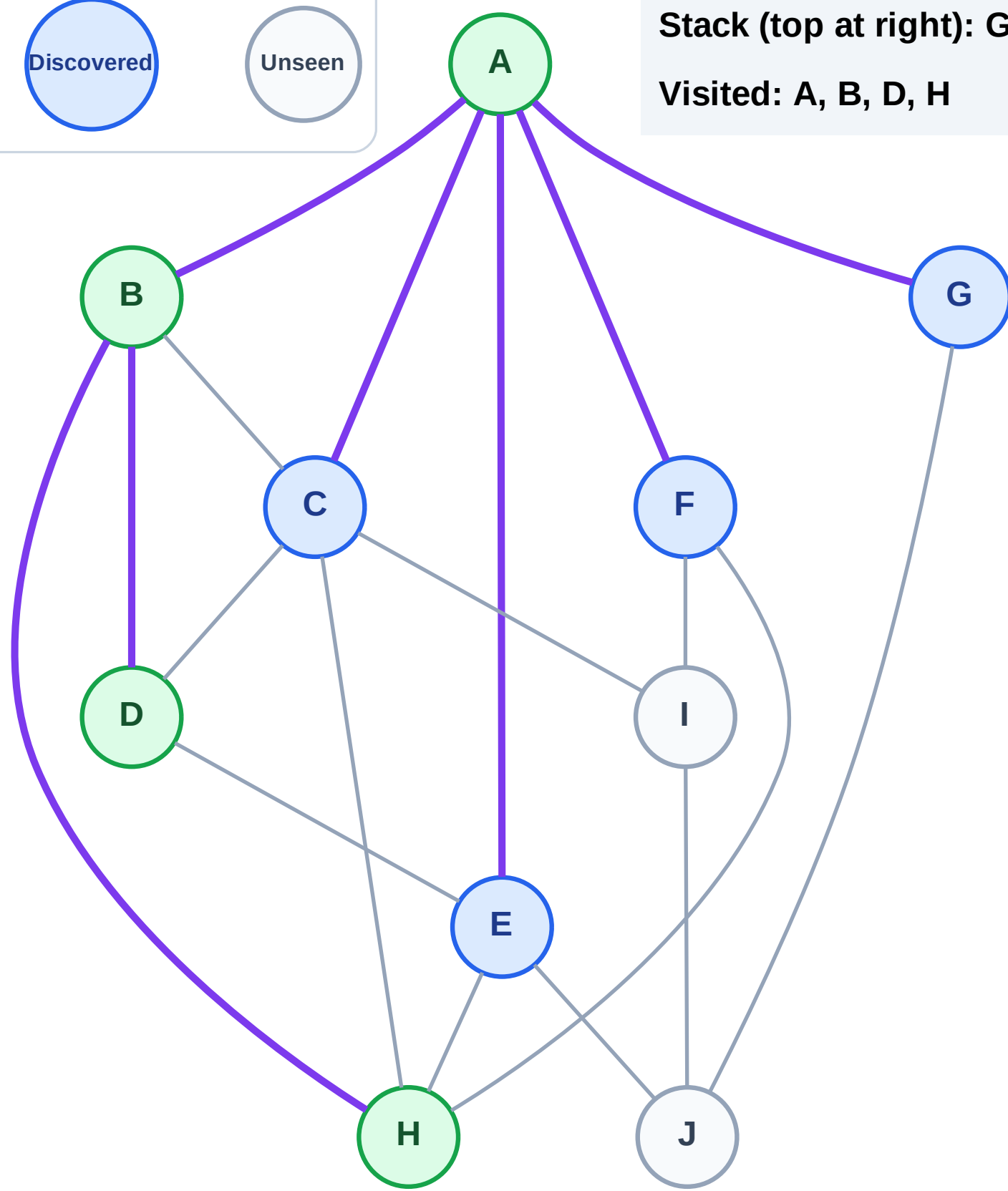
Step 9: No new neighbors from H

Legend



Stack (top at right): G | F | E | C

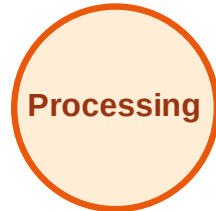
Visited: A, B, D, H



DFS Traversal

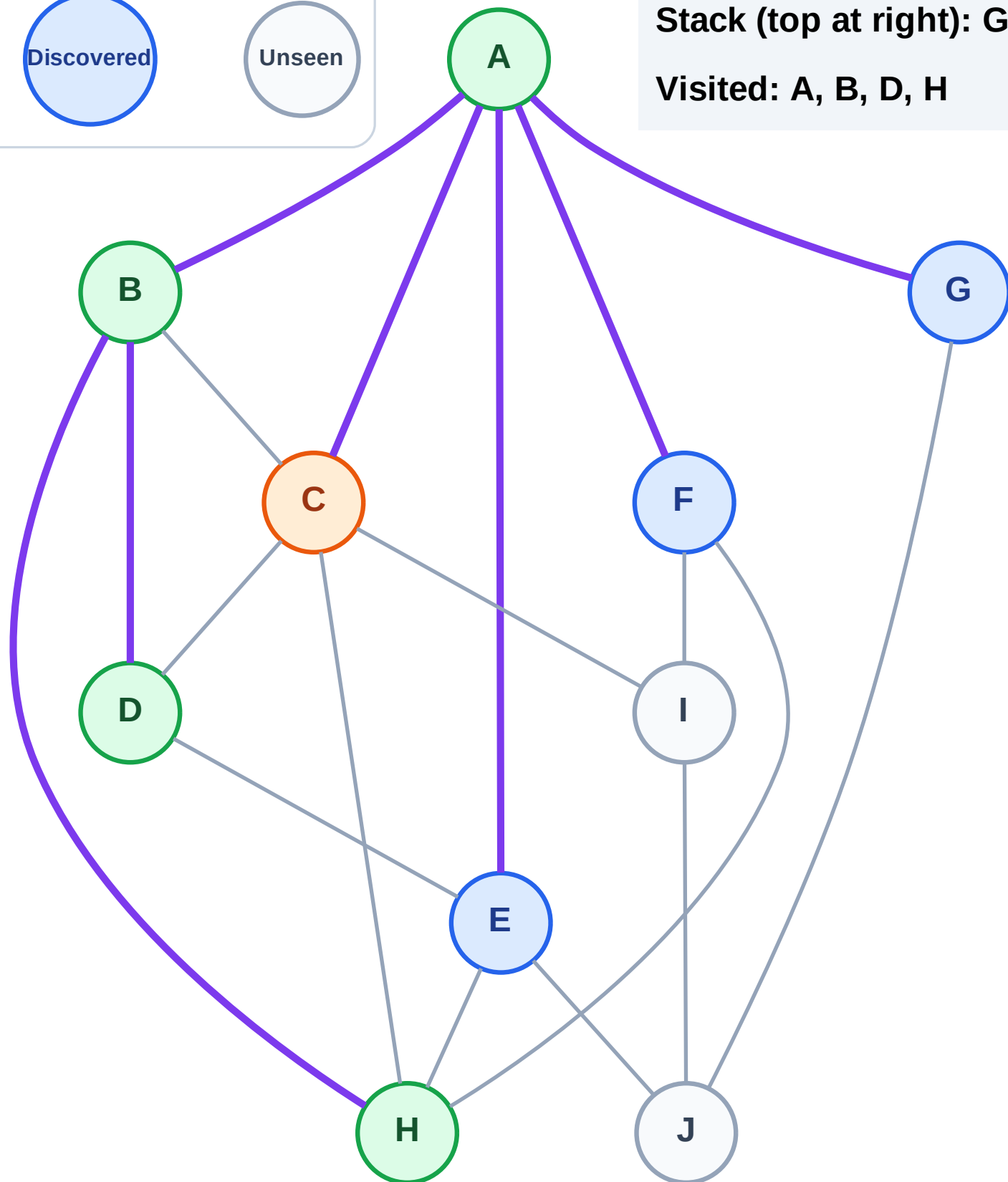
Step 10: Process node C

Legend



Stack (top at right): G | F | E

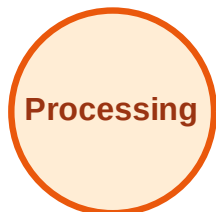
Visited: A, B, D, H



DFS Traversal

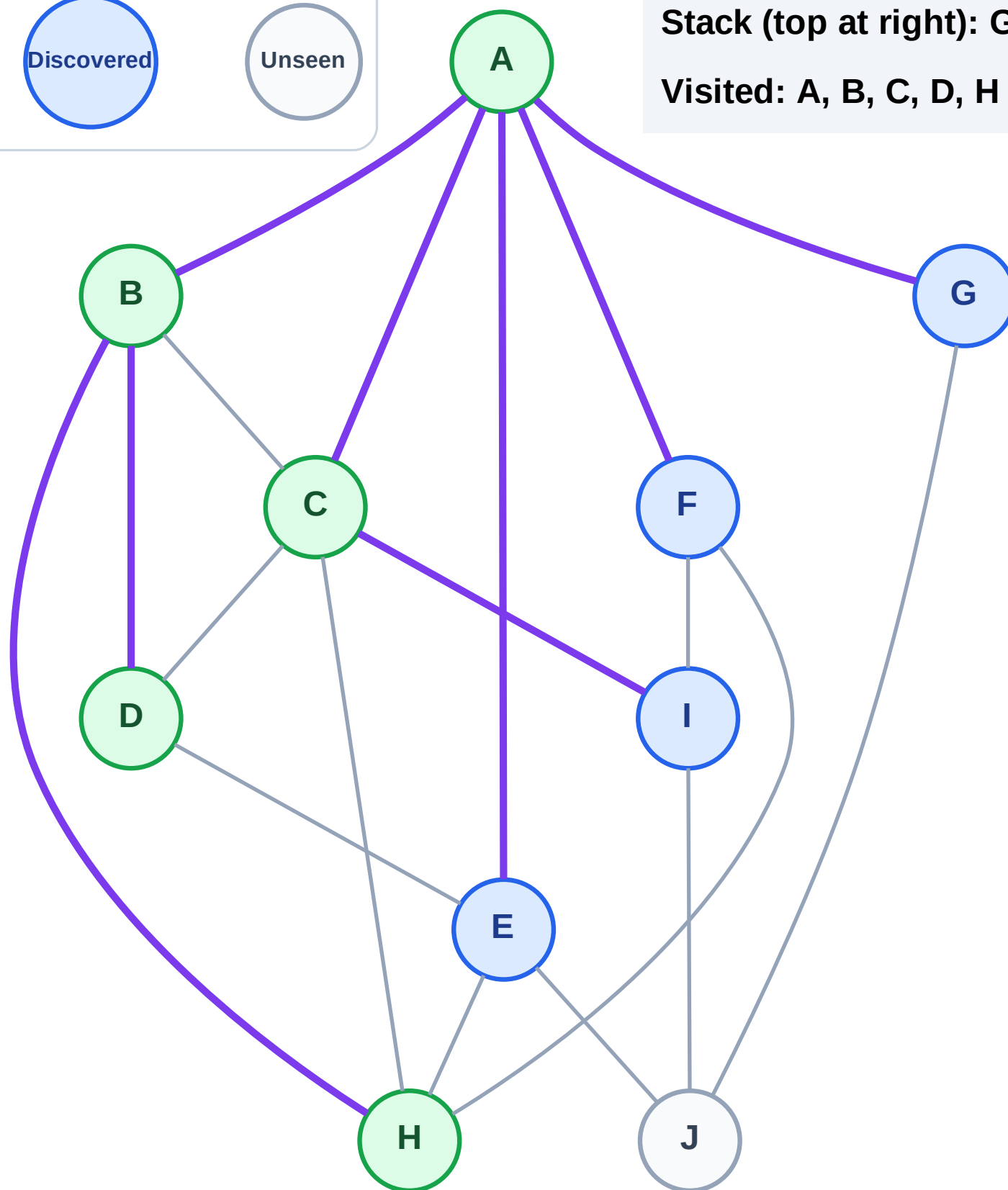
Step 11: Discover I from C

Legend



Stack (top at right): G | F | E | I

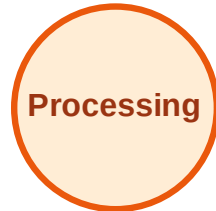
Visited: A, B, C, D, H



DFS Traversal

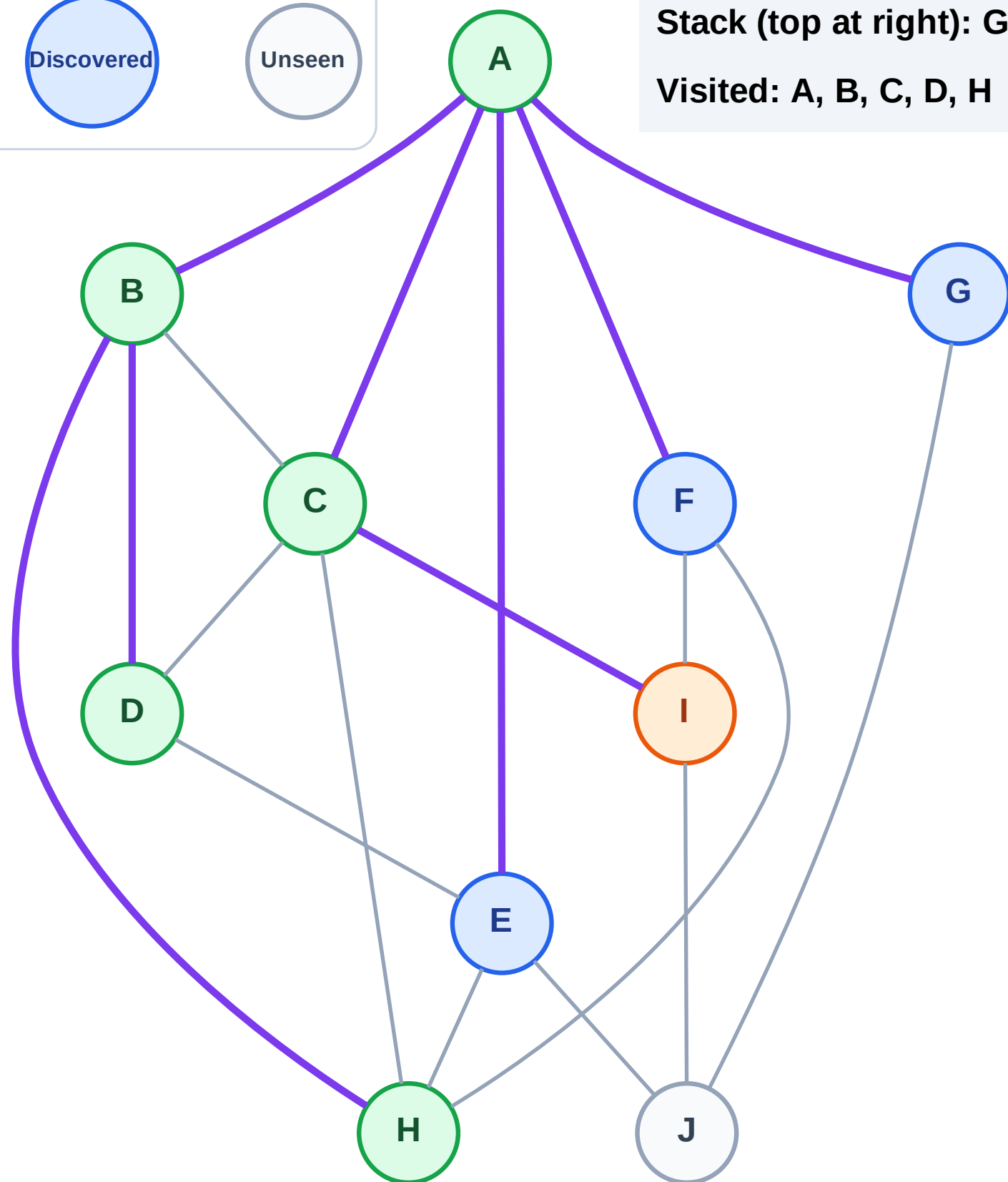
Step 12: Process node I

Legend



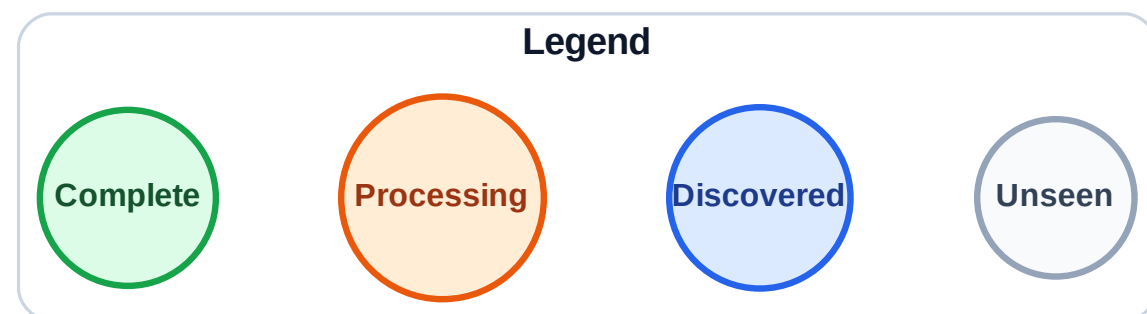
Stack (top at right): G | F | E

Visited: A, B, C, D, H

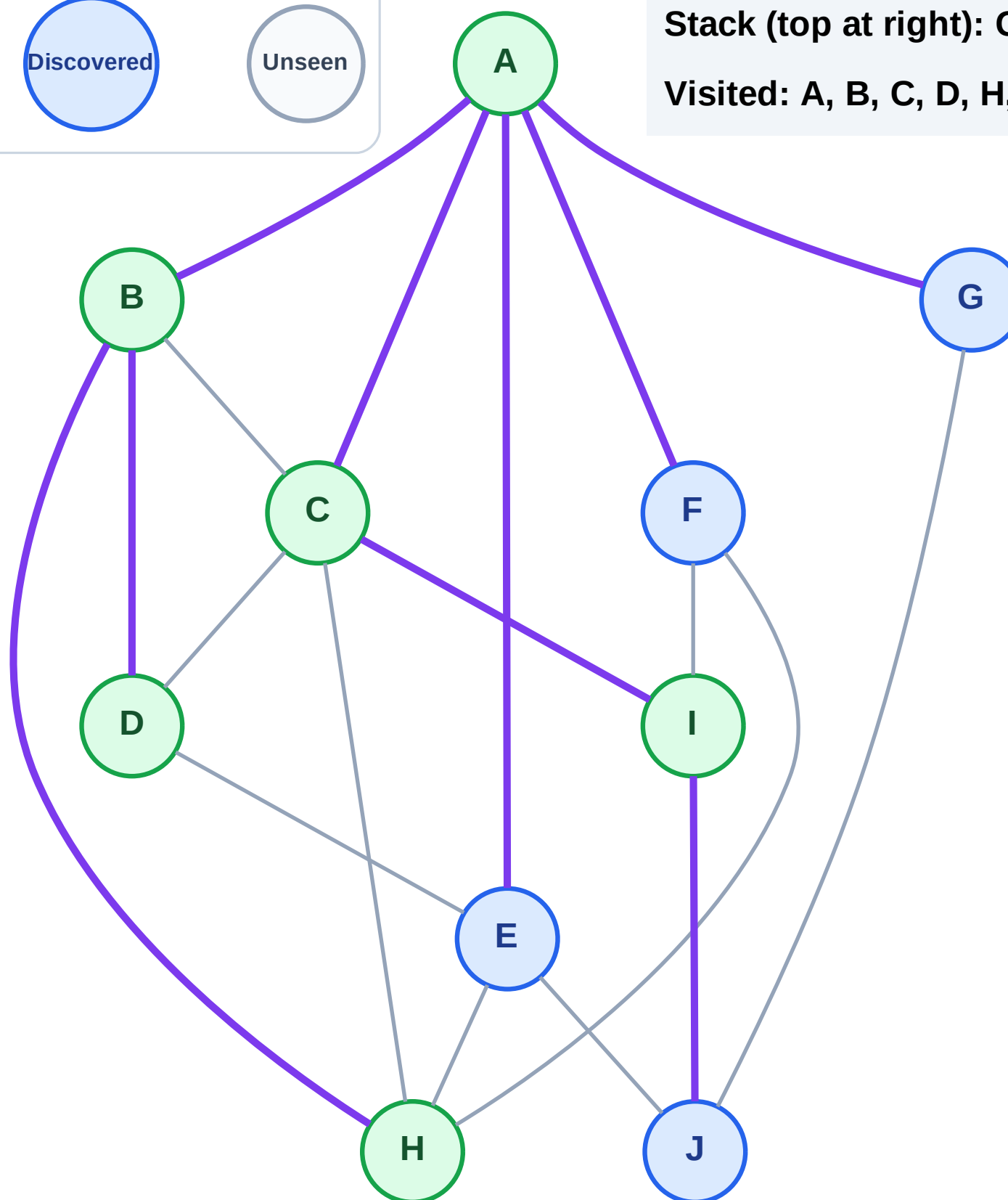


DFS Traversal

Step 13: Discover J from I



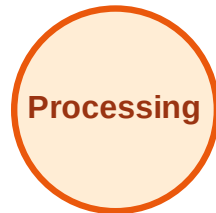
Stack (top at right): G | F | E | J
Visited: A, B, C, D, H, I



DFS Traversal

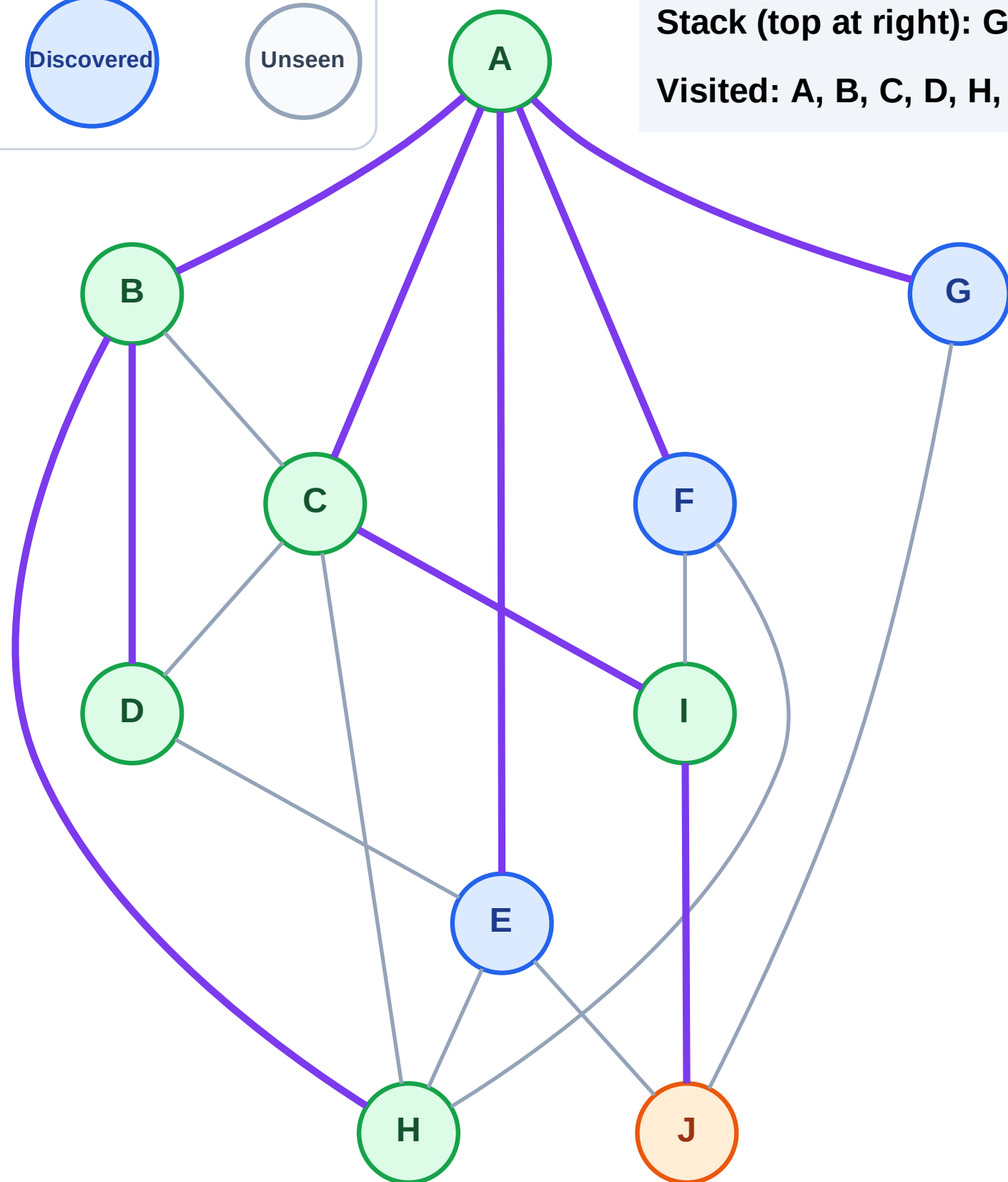
Step 14: Process node J

Legend



Stack (top at right): G | F | E

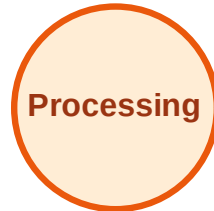
Visited: A, B, C, D, H, I



DFS Traversal

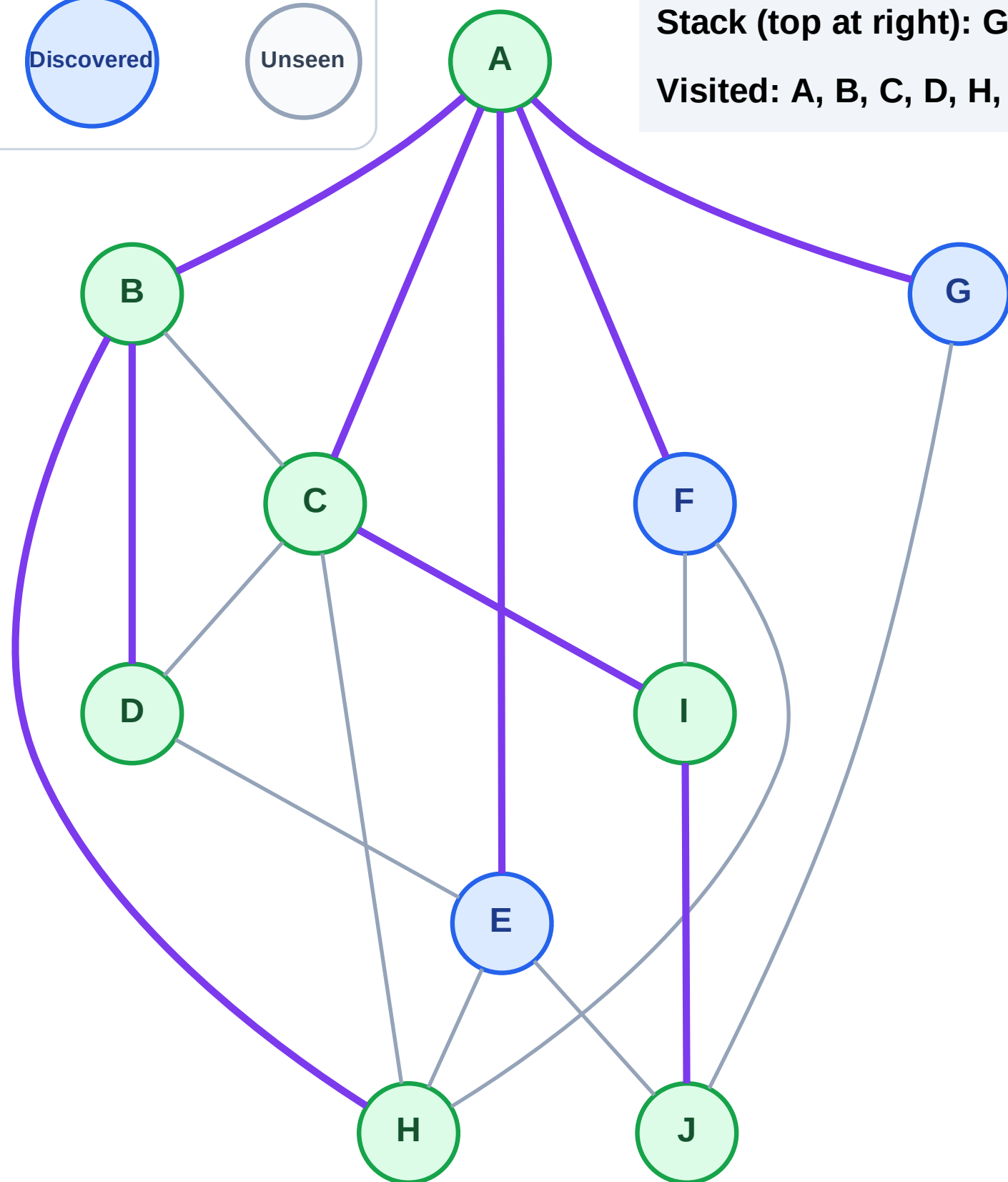
Step 15: No new neighbors from J

Legend



Stack (top at right): G | F | E

Visited: A, B, C, D, H, I, J



DFS Traversal

Step 16: Process node E

Legend

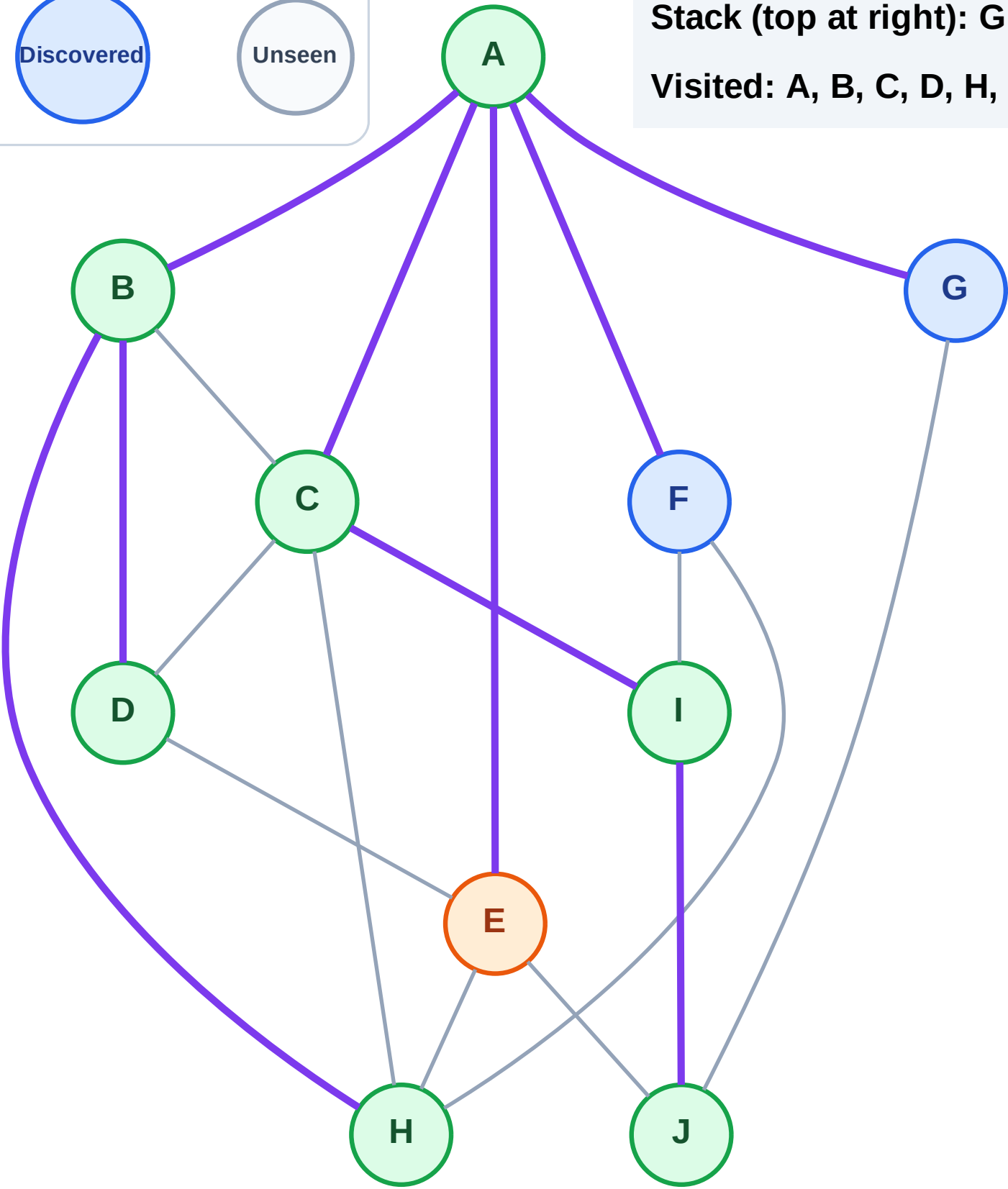
Complete

Processing

Discovered

Unseen

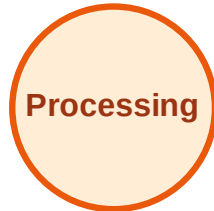
Stack (top at right): G | F
Visited: A, B, C, D, H, I, J



DFS Traversal

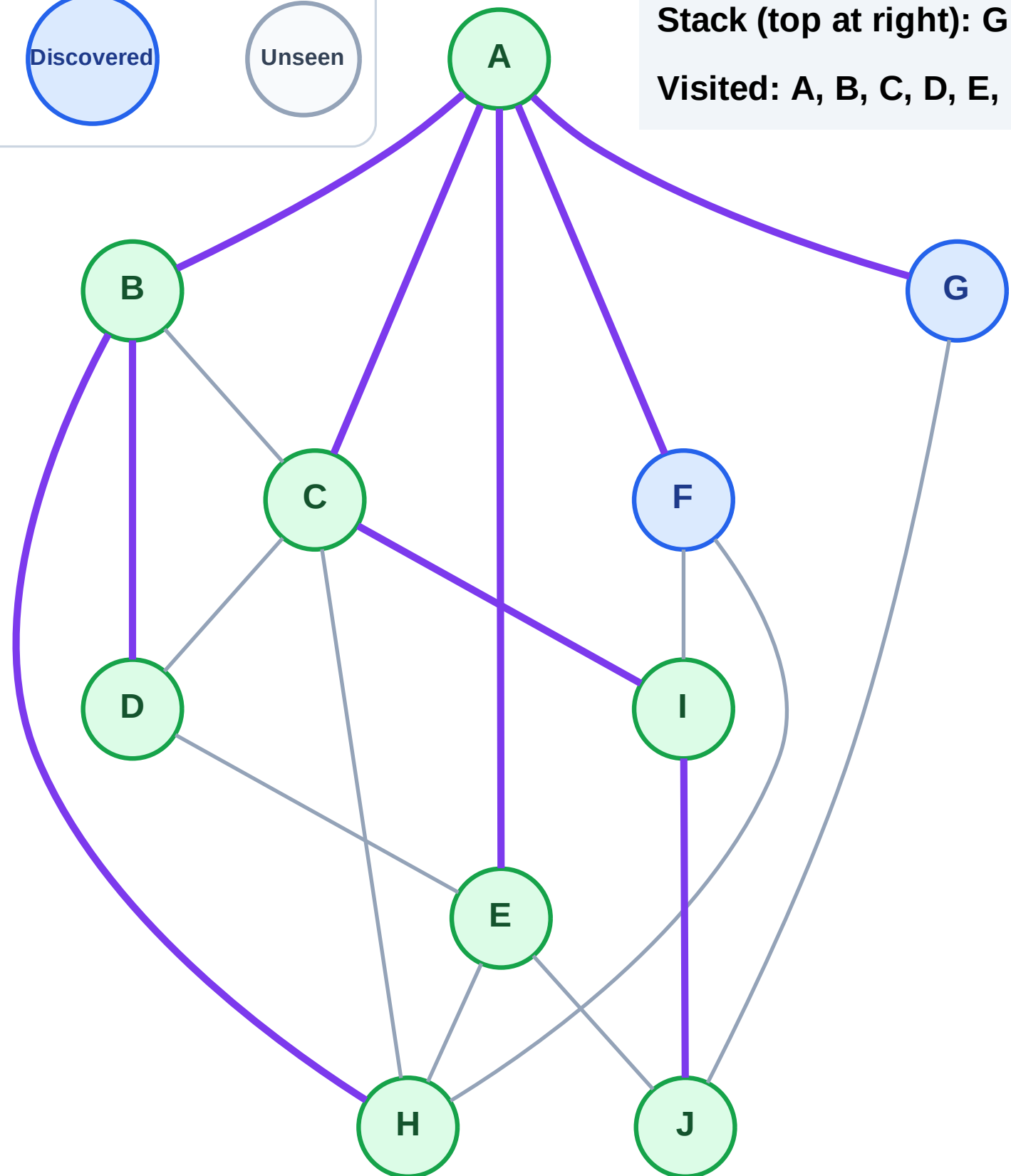
Step 17: No new neighbors from E

Legend



Stack (top at right): G | F

Visited: A, B, C, D, E, H, I, J



DFS Traversal

Step 18: Process node F

Legend

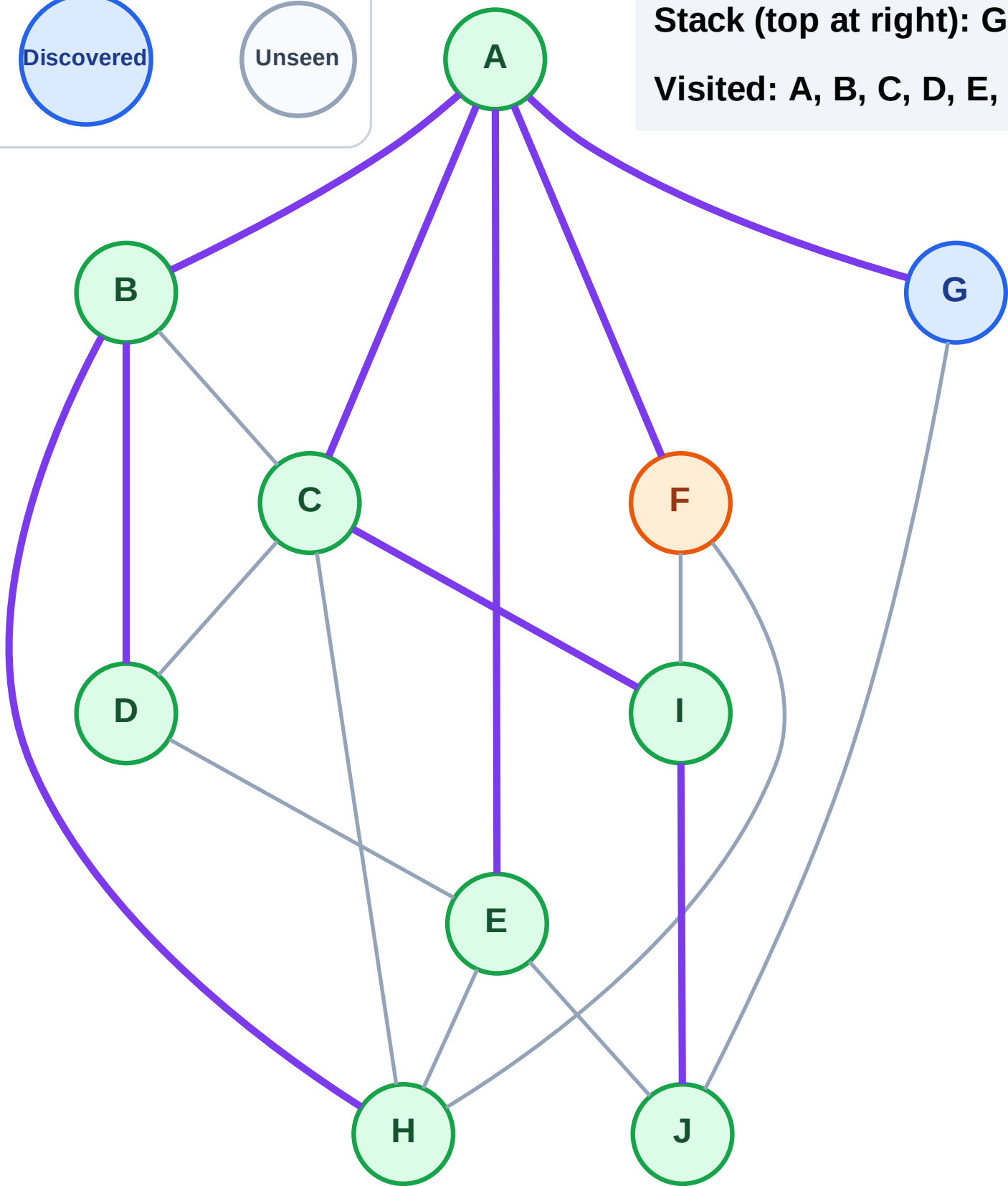
Complete

Processing

Discovered

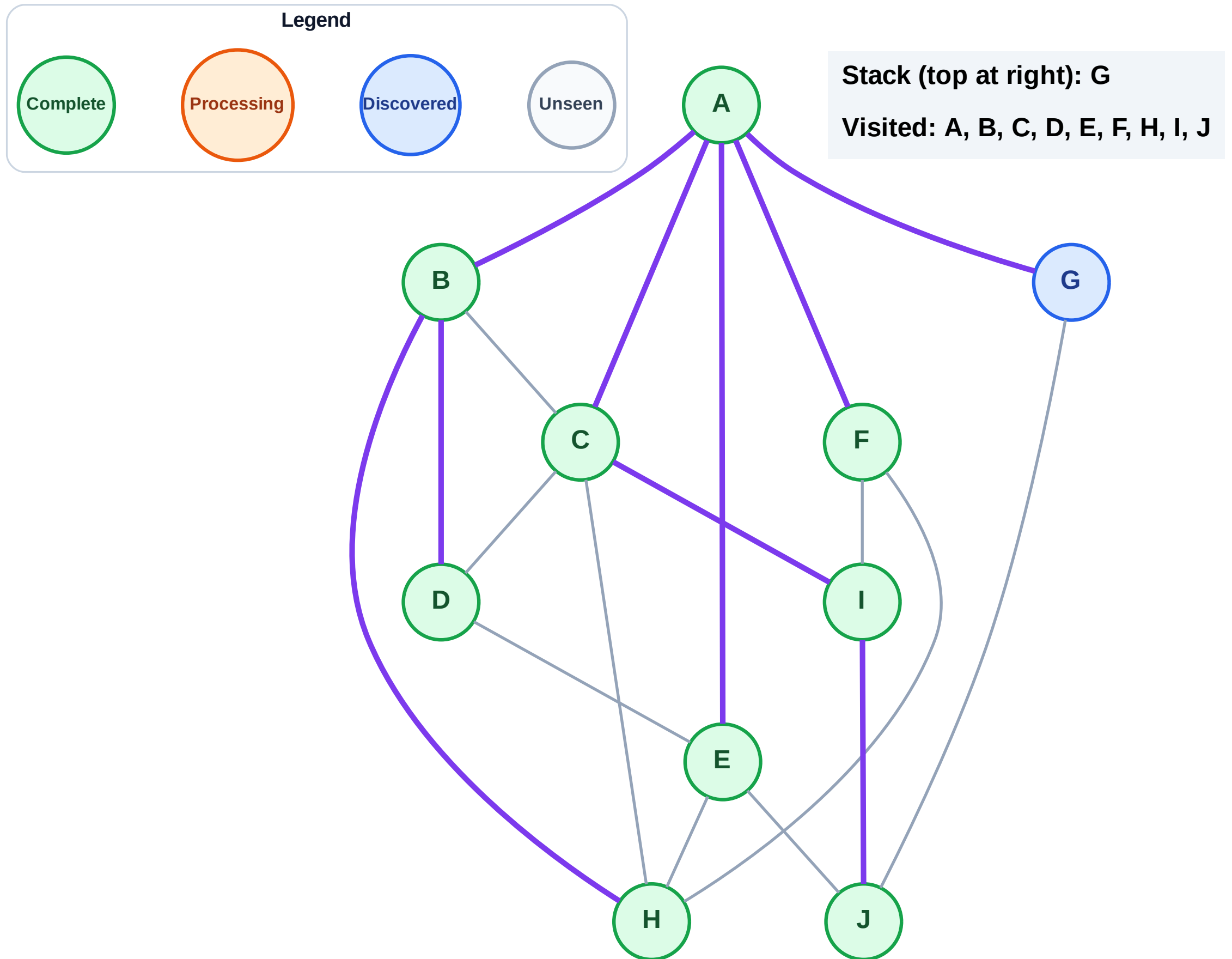
Unseen

Stack (top at right): G
Visited: A, B, C, D, E, H, I, J



Step 19: No new neighbors from F

Step 19: No new neighbors from F



DFS Traversal

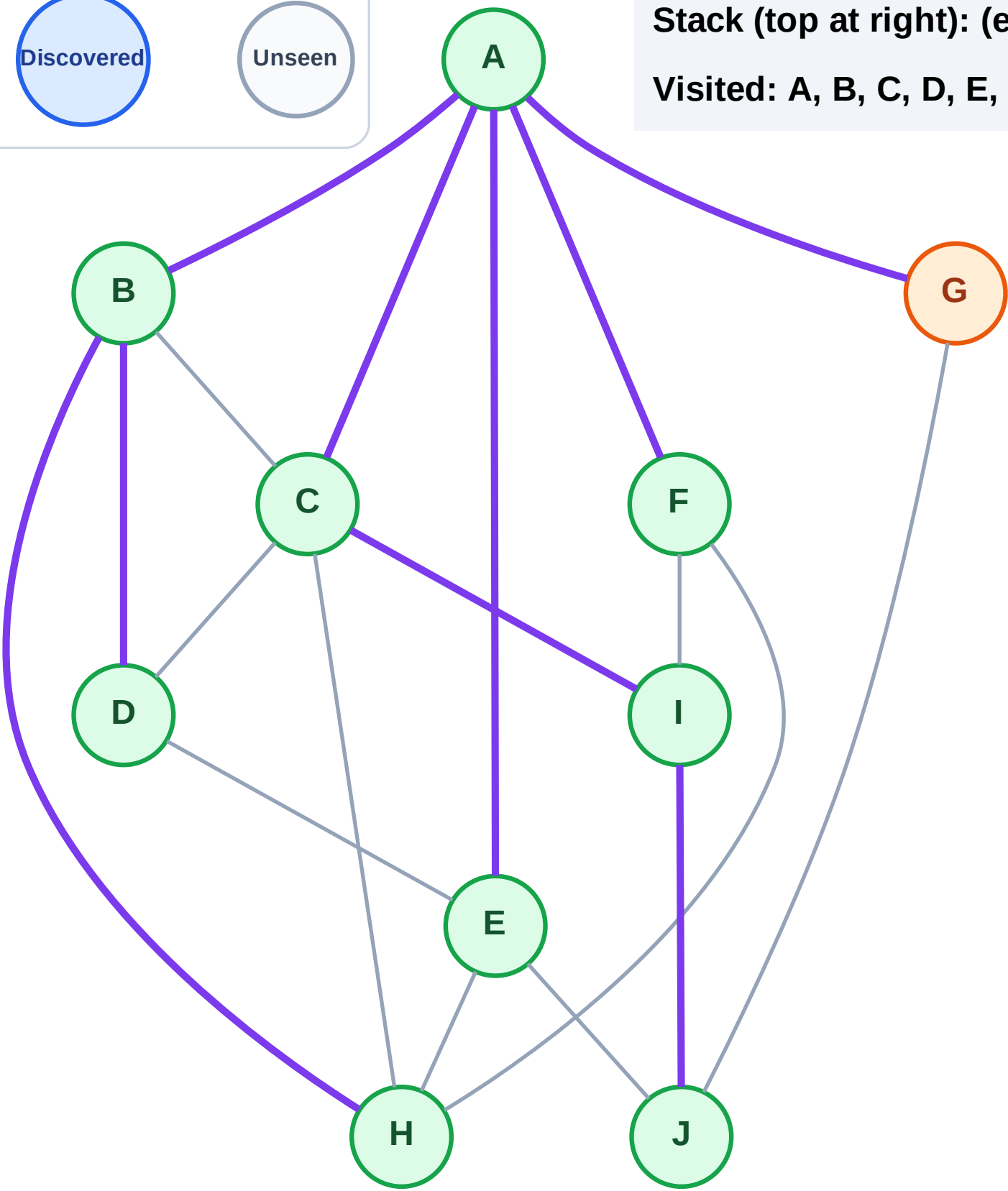
Step 20: Process node G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

